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Median-of-Means Instrumental Variable Regression

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1 Introduction

The just-identified instrumental variable (IV) estimator is a ratio of two sample means. This makes it fragile in two distinct ways: the numerator is a sample mean of a possibly heavy-tailed product, and the denominator can be close to zero. The effects of large outliers on IV estimation was studied by Young (2022), where 1309 IV regressions in 30 papers were examined and it was shown that high leverage and non-i.i.d. errors are pervasive. This distorts IV tests and rarely rejects the OLS estimate. Additionally, F-statistics based on pre-tests become largely uninformative about size and bias.

In the presence of heavy tails, the sample mean is no longer an efficient measure of central tendency. As such, the quality of standard regression estimation techniques, such as OLS, can degrade significantly when this is the case. Under only a finite-variance assumption, the empirical mean’s deviation from the true mean scales as $|\hat{\mu} - \mu| \lesssim \frac{1}{\sqrt{n\delta}}$ with confidence $1 - \delta$, whereas sub-Gaussian estimators achieve the exponentially better rate $|\hat{\mu} - \mu| \lesssim \sqrt{\ln(1/\delta)/n}$.

discuss about estimators that solve this and cite Devroye, Lerasle, Lugosi and Oliveira (2016)

A technique that has gained traction in the machine learning literature is the Median-of-Means (MoM) estimator (Lecué & Lerasle, 2020), which attains sub-Gaussian deviation bounds under nothing more than a finite variance. The foundational studies of this method were given by Nemirovskij and Yudin (1983), Jerrum, Valiant and Vazirani (1986), and Alon, Matias and Szegedy (1996). A comprehensive survey and analysis of the MoM estimator was given by Lugosi and Mendelson (2019)

MoM has been carried into OLS estimation, but not into instrumental variable estimation; MoM is defined for a mean, whereas the IV estimand is a ratio of means. Since there is no canonical way to take a median of a ratio, novel methods have to be constructed that, not only estimate it, but also give the most preferable theoretical guarantees, and empirical results.

We propose and analyse two MoM-based IV estimators. The Ratio-of-Medians (RoM) estimator takes the coordinate-wise median of the block cross-moments and then divides; the Median-of-Ratios (MoR) estimator divides within each block and then takes the median of the block-level IV estimates. Both of these estimators are Sub-Gaussian, but the MoR consistently shows better empirical results throughout the simulations.

The robustness is not free. In standard mean estimation MoM costs constants, but in ratio estimation it costs a per-block sample size floor. That is, in order to guarantee a Sub-Gaussian estimator with a certain probability, a minimum per-block sample size is needed. MoM-type IV estimators pay for tail robustness by spending instrument strength.

In order to be able to conduct inference under the MoM framework, we construct a median-of-means Anderson-Rubin (MoM-AR) test whose size is bounded by δ in finite samples under the same assumptions, and show that the confidence set obtained by inverting it can be computed exactly in closed form from at most $2k$ candidate endpoints. This test depends on an unknown

scale parameter σ that has to be estimated. We therefore construct a self-normalised Anderson-Rubin test which cancels the unknown σ instead of estimating it or treating it as known.

discuss size and power costs of tests

We evaluate the estimators and tests on simulated data across Gaussian, t , and Pareto shocks and a sweep of instrument strengths. Under light tails the MoM estimators cost what the theory says they should. Under $t(2.1)$ errors the Median-of-Ratios estimator improves on the sample mean in the tails and matches Catoni's estimator. Two costs appear that the theory does not price. First, the MoM-AR threshold is roughly seven times the true scale of the statistic under the null. Second, the random partition contributes a third or more of the total variance of the estimates once the instrument is weak.

We then apply the same procedures to Angrist and Krueger (1991) and Angrist and Evans (1998), chosen because the strength conditions fail in the first and hold in the second. Both Wald estimates are reproduced exactly. The MoM-AR sets are uninformative in both applications, the self-normalised test is the one that can be recommended, and the partition changes the reported estimate in the weak application while leaving every verdict unchanged in the strong one.

Finally, the constituent steps of every theorem is machine checked in Lean 4 using the Mathlib library, so that the measure-theoretic and probabilistic foundations are inherited rather than assumed.

All theorems except those that are aggregated, machine check the aggregated theorems

2 Preliminaries

2.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \tag{1}$$

where Y_i is the variable of interest, X_i is an endogenous regressor, and Z_i is the instrumental variable (IV). Additionally, $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term.

For any random quantity f , we write $\mu_f := \mathbb{E}[f]$ and $\sigma_f^2 := \text{Var}(f)$ for its mean and variance. All population quantities below are instances of this scheme; for example, $\mu_X = \mathbb{E}[X_i]$, $\mu_{ZX} = \mathbb{E}[Z_i X_i]$, and $\sigma_{ZX}^2 = \text{Var}(Z_i X_i)$.

We impose the following conditions.

(A1) **Exogeneity.** $\mu_{Z\varepsilon} = 0$.

(A2) **Relevance.** $\mu_{ZX} \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZX}^2, \sigma_{ZY}^2, \sigma_{Z\varepsilon}^2 < \infty$.

By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY}/\mu_{ZX}$.

Remark 2.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i , X_i , Z_i , or ε_i .*

Remark 2.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$.

2.2 Sub-Gaussian Estimators

Definition 2.3 (Sub-Gaussian Estimators). *Let $(X_i)_{i=1}^n$ be a random sample with mean μ_X and variance σ_X^2 , and let $\hat{\mu}_X$ denote an estimator for μ_X . For some constant $L > 0$, $\hat{\mu}_X$ is said to be a sub-Gaussian estimator for μ_X for the given confidence level δ if*

$$\mathbb{P} \left[|\hat{\mu}_X - \mu_X| > L \sigma_X \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

2.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye et al. (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}_{\text{MoM}}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute the block mean $\hat{\mu}_X^{(j)} = \frac{1}{m} \sum_{i \in B_j} X_i$;

return $\hat{\mu}_{\text{MoM}} = \text{med}(\hat{\mu}_X^{(1)}, \dots, \hat{\mu}_X^{(k)})$;

Theorem 2.4 (MoM concentration). *Let $(X_i)_{i=1}^n$ be i.i.d. with mean μ_X and variance $\sigma_X^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\text{MoM}}$ the MoM estimator of μ_X . Then*

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}} \right] \leq e^{-k/8}. \quad (4)$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0, 1)$, this becomes

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \sigma_X \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta, \quad (5)$$

which is sub-Gaussian in the sense of definition 2.3 with $L = \sqrt{32}$.

Wrong. with $k = \lceil 8 \ln(1/\delta) \rceil$ the radius $2\sigma_X/\sqrt{m}$ exceeds $\sigma_X \sqrt{32 \ln(1/\delta)/n}$.

All of the following theorems assume $n = km$

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}^{(j)}] = \mu_X$ and $\text{Var}(\bar{X}^{(j)}) = \sigma_X^2/m$. By Chebyshev's inequality with threshold $2\sigma_X/\sqrt{m}$,

$$\mathbb{P}\left[\left|\bar{X}^{(j)} - \mu_X\right| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \frac{\sigma_X^2/m}{4\sigma_X^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta^{(j)} = \mathbf{1}\{|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta^{(j)}] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta^{(j)})_{j=1}^k$ are independent.

If $|\hat{\mu}_{\text{MoM}} - \mu_X| > 2\sigma_X/\sqrt{m}$, the median exceeds $\mu_X + 2\sigma_X/\sqrt{m}$ or falls below $\mu_X - 2\sigma_X/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P}\left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} \geq \frac{k}{2}\right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta^{(j)}$ consists of k independent random variables with $\zeta^{(j)} \in [0, 1]$ and $\mathbb{E}[\sum_j \zeta^{(j)}] \leq k/4$. For $\sum_j \zeta^{(j)}$ to reach $k/2$, it must exceed its mean by at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} - \mathbb{E}\left[\sum_j \zeta^{(j)}\right] \geq \frac{k}{4}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$. $\square$

3 The Standard Instrumental Variable Estimator

Definition 3.1 (Standard IV estimator). *Consider the structural equation (1). The standard IV estimator $\hat{\beta}_{\text{IV}}$ is defined by*

$$\hat{\beta}_{\text{IV}} = \frac{\hat{\mu}_{ZY}}{\hat{\mu}_{ZX}},$$

Lemma 3.2 (Error decomposition). *Under (1) and (A1),*

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\hat{\mu}_{Z\varepsilon}}{\hat{\mu}_{ZX}}, \tag{6}$$

where $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\hat{\mu}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i(\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX} + \hat{\mu}_{Z\varepsilon}.$$

Rearranging and dividing by $\hat{\mu}_{ZX}$ gives $\hat{\beta}_{\text{IV}} = \beta + \hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$, which is (6). By (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ by linearity of expectation. $\square$

Theorem 3.3 (Standard IV concentration). *Assume (A1)–(A3). For any $\delta \in (0, 1)$, if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad (7)$$

then

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}\right] \leq \delta. \quad (8)$$

Proof. By Lemma 3.2, the estimation error equals the ratio $\hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$. To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

Step 1 (Two-event decomposition). For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A(D) = \{|\hat{\mu}_{ZX}| \geq D\}, \quad B(D, t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}.$$

On $A(D) \cap B(D, t)$, the denominator satisfies $|\hat{\mu}_{ZX}| \geq D$ by definition of $A(D)$, so

$$\left|\hat{\beta}_{IV} - \beta\right| = \frac{|\hat{\mu}_{Z\varepsilon}|}{|\hat{\mu}_{ZX}|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D, t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A^c = \{|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2\}$ implies a large deviation of $\hat{\mu}_{ZX}$ from its mean. Specifically, if $|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$|\hat{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - |\hat{\mu}_{ZX}| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A^c \subseteq \{|\hat{\mu}_{ZX} - \mu_{ZX}| > |\mu_{ZX}|/2\}$. Since $\text{Var}(\hat{\mu}_{ZX}) = \sigma_{ZX}^2/n$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A^c] \leq \frac{\sigma_{ZX}^2/n}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2}.$$

The failure event is $B^c = \{|\hat{\mu}_{Z\varepsilon}| > t|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ and $\text{Var}(\hat{\mu}_{Z\varepsilon}) = \sigma_{Z\varepsilon}^2/n$, Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2}.$$

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most δ , and we allocate half the budget to each event.

Budget for A^c . Setting $\mathbb{P}[A^c] \leq \delta/2$ requires

$$\frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2},$$

which is exactly condition (7).

Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \quad \square$$

Remark 3.4 (Polynomial dependence on δ). *The bound (8) contains the factor $1/\sqrt{\delta}$, which is polynomial in $1/\delta$. Catoni (2012) show that this is unavoidable, since for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the $1/\sqrt{\delta}$ rate cannot be improved without additional distributional assumptions. The MoM-based estimators in Sections 4–5 circumvent this by operating on block means.*

Remark 3.5 (The symmetric threshold is not optimal). *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between 0 and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ . Section 3.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from 8 to 1 and improves the error constant by a factor of $2\sqrt{2}$.*

3.1 Optimised Threshold

The proof of Theorem 3.3 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

Theorem 3.6 (Standard IV concentration, optimised threshold). *Assume (A1)–(A3). Define*

$$\rho_{\text{IV}} = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0, 1)$, if $\rho_{\text{IV}} < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), then the choice $D^ = (1 - \rho_{\text{IV}}^{1/3})|\mu_{ZX}|$ is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1 - \rho_{\text{IV}}^{1/3})^{3/2}}\right] \leq \delta. \quad (9)$$

Proof. Steps 1 and 2 are identical to those of Theorem 3.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2 D^2} \leq \delta. \quad (10)$$

Step 3 (Optimise D , then solve for t). Write $D = \eta |\mu_{ZX}|$ with $\eta \in (0, 1)$. Dividing (10) through by δ and substituting $\rho_{IV} = \sigma_{ZX}^2 / (\delta n \mu_{ZX}^2)$:

$$\frac{\rho_{IV}}{(1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \leq 1.$$

Solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \quad f(\eta) = \eta^2 \left(1 - \frac{\rho_{IV}}{(1 - \eta)^2} \right).$$

Minimising t is equivalent to maximising f over $\eta \in (0, 1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta \left(1 - \frac{\rho_{IV}}{(1 - \eta)^2} \right) - \frac{2\rho_{IV}\eta^2}{(1 - \eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho_{IV}}{(1 - \eta)^2} - \frac{\rho_{IV}\eta}{(1 - \eta)^3} = 0 \iff (1 - \eta)^3 - \rho_{IV}(1 - \eta) - \rho_{IV}\eta = 0 \iff (1 - \eta)^3 = \rho_{IV}.$$

Hence $\eta^* = 1 - \rho_{IV}^{1/3}$, which lies in $(0, 1)$ if and only if $\rho_{IV} < 1$, the stated instrument strength condition. Substituting $(1 - \eta^*)^2 = \rho_{IV}^{2/3}$:

$$f(\eta^*) = (1 - \rho_{IV}^{1/3})^2 \left(1 - \frac{\rho_{IV}}{\rho_{IV}^{2/3}} \right) = (1 - \rho_{IV}^{1/3})^2 (1 - \rho_{IV}^{1/3}) = (1 - \rho_{IV}^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho_{IV}^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P} \left[\left| \hat{\beta}_{IV} - \beta \right| > t^* \right] \leq \delta$. □

Remark 3.7 (Comparison with the symmetric bound). As $\rho_{IV} \rightarrow 0$ (strong instruments or large n), the correction factor $(1 - \rho_{IV}^{1/3})^{-3/2} \rightarrow 1$ and (9) approaches $\sigma_{Z\varepsilon} / (|\mu_{ZX}| \sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho_{IV} \rightarrow 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 3.3: $\rho_{IV} < 1$ versus $\rho_{IV} < 1/8$.

3.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\hat{\mu}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\hat{\mu}_{ZX}$ from μ_{ZX} causes

the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 3.8 (Cantelli's inequality). *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\text{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$.

Theorem 3.9 (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{\text{IV}} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$ and define*

$$h(\eta) = \eta^2 \frac{\delta(\gamma_{\text{IV}} + (1 - \eta)^2) - \gamma_{\text{IV}}}{\gamma_{\text{IV}} + (1 - \eta)^2}, \quad \eta \in (0, 1). \quad (11)$$

If

$$n > \frac{(1 - \delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{\text{IV}} < \frac{\delta}{1 - \delta}, \quad (12)$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta_{\text{IV}}^*)}}\right] \leq \delta, \quad (13)$$

where η_{IV}^ is the maximiser of h over its feasible domain.*

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 3.3, define $A(D) = \{\hat{\mu}_{ZX} \geq D\}$ and $B(D, t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}$. On $A(D) \cap B(D, t)$, $|\hat{\beta}_{\text{IV}} - \beta| \leq t$.

Step 2 (Cantelli on A^c , Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\hat{\mu}_{ZX} < D\} \subseteq \{\hat{\mu}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 3.8 applies with $W = \hat{\mu}_{ZX}$, $\sigma^2 = \sigma_{ZX}^2/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{\sigma_{ZX}^2/n + (\mu_{ZX} - D)^2}.$$

The event $B(D, t)^c = \{|\hat{\mu}_{Z\varepsilon}| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Step 3 (Constraint and optimisation). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{\text{IV}} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$:

$$\frac{\gamma_{\text{IV}}}{\gamma_{\text{IV}} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2\eta^2\mu_{ZX}^2} \leq \delta.$$

Isolating the second term and solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{n\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (11). The function $h(\eta)$ is positive only when $\delta(\gamma_{IV} + (1 - \eta)^2) > \gamma_{IV}$, i.e., $(1 - \eta)^2 > \gamma_{IV}(1 - \delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{IV} < \delta/(1 - \delta)$, which is condition (12). Minimising t over the feasible region by maximising h and applying the union bound gives (13). $\square$

Remark 3.10 (When Cantelli helps). *Compared with the optimised Chebyshev condition ($n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), the Cantelli condition (12) is weaker by a factor of $1/(1 - \delta)$. At $\delta = 0.05$ this ratio is $1/0.95 \approx 1.05$, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at $\delta = 0.5$ the factor reaches 2. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section 5.2), Cantelli extends the valid range of the instrument strength parameter from $\gamma < 1/4$ to $\gamma < 1/3$, a qualitatively more substantial gain.*

Remark 3.11 (No tractable closed form for η_{IV}^*). *The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no tractable closed-form solution. The optimised Chebyshev version (Theorem 3.6) yielded the clean form $\eta^* = 1 - \rho_{IV}^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound (13) must therefore be evaluated numerically for specific parameter values.*

4 The Ratio-of-Medians (RoM) Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{RM}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute the block means $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;

Compute the coordinate-wise medians $\tilde{\mu}_{ZY} = \text{med}(\hat{\mu}_{ZY}^{(1)}, \dots, \hat{\mu}_{ZY}^{(k)})$ and

$\tilde{\mu}_{ZX} = \text{med}(\hat{\mu}_{ZX}^{(1)}, \dots, \hat{\mu}_{ZX}^{(k)})$;

return $\hat{\beta}_{RM} = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}}$;

Theorem 4.1 (RoM concentration). *Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(2/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \tag{14}$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{RM} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \tag{15}$$

Proof. The proof applies the scalar MoM result (Theorem 2.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 2.4 applied to the sequences $(Z_i Y_i)_{i=1}^n$ and $(Z_i X_i)_{i=1}^n$,

$$\mathbb{P}\left[|\tilde{\mu}_{ZY} - \mu_{ZY}| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \leq e^{-k/8}, \quad \mathbb{P}\left[|\tilde{\mu}_{ZX} - \mu_{ZX}| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \leq e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \quad B = \left\{|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \leq e^{-k/8}$ and $\mathbb{P}[B^c] \leq e^{-k/8}$. Note that A and B are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq 2e^{-k/8}.$$

Setting $k = \lceil 8 \ln(2/\delta) \rceil$ ensures $2e^{-k/8} \leq \delta$.

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\text{RM}} - \beta = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX}}{\tilde{\mu}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX} = \mu_{ZX}[(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\text{RM}} - \beta = \frac{(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})}{\tilde{\mu}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{|\tilde{\mu}_{ZY} - \mu_{ZY}| + |\beta| |\tilde{\mu}_{ZX} - \mu_{ZX}|}{|\tilde{\mu}_{ZX}|}.$$

Step 4 (Substitute bounds). On event A , $|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq 2\sigma_{ZY}/\sqrt{m}$. On event B , $|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$|\tilde{\mu}_{ZX}| \geq |\mu_{ZX}| - |\tilde{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta| \sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \leq \delta$, the result follows. $\square$

5 The Median-of-Ratios (MoR) Estimator

Algorithm 3: Median-of-Ratios Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\text{MR}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute the block means $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
 Compute block level IV estimates $\hat{\beta}^{(j)} = \frac{\hat{\mu}_{ZY}^{(j)}}{\hat{\mu}_{ZX}^{(j)}};$

return $\hat{\beta}_{\text{MR}} = \text{median}(\hat{\beta}^{(1)}, \dots, \hat{\beta}^{(k)})$;

Lemma 5.1 (Block-level error decomposition). *Under (1) and (A1), for each block j ,*

$$\hat{\beta}^{(j)} - \beta = \frac{\bar{\varepsilon}_j^Z}{\hat{\mu}_{ZX}^{(j)}}, \quad (16)$$

where $\bar{\varepsilon}_j^Z = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i$ satisfies $\mathbb{E}[\bar{\varepsilon}_j^Z] = 0$.

Proof. Multiply $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average over block B_j :

$$\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i (\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX}^{(j)} + \bar{\varepsilon}_j^Z.$$

Dividing by $\hat{\mu}_{ZX}^{(j)}$ gives $\hat{\beta}^{(j)} = \beta + \bar{\varepsilon}_j^Z / \hat{\mu}_{ZX}^{(j)}$. By (A1) and linearity of expectation, $\mathbb{E}[\bar{\varepsilon}_j^Z] = \frac{1}{m} \sum_{i \in B_j} \mathbb{E}[Z_i \varepsilon_i] = 0$. $\square$

Theorem 5.2 (MoR concentration). *Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(1/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (17)$$

holds, then

$$\mathbb{P} \left[\left| \hat{\beta}_{\text{MR}} - \beta \right| > \frac{4\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{m}} \right] \leq \delta. \quad (18)$$

Proof. The proof applies the MoM counting argument directly to the block-level IV estimates $\hat{\beta}^{(j)}$.

Step 1 (Two-event decomposition). By Lemma 5.1, $\left| \hat{\beta}^{(j)} - \beta \right| = \left| \bar{\varepsilon}_j^Z \right| / \left| \hat{\mu}_{ZX}^{(j)} \right|$. To bound this ratio, we control the numerator above and the denominator below. For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A_j(D) = \left\{ \left| \hat{\mu}_{ZX}^{(j)} \right| \geq D \right\}, \quad B_j(D, t) = \left\{ \left| \bar{\varepsilon}_j^Z \right| \leq tD \right\}.$$

On $A_j(D) \cap B_j(D, t)$, the denominator satisfies $|\hat{\mu}_{ZX}^{(j)}| \geq D$ by definition of $A_j(D)$, so

$$|\hat{\beta}^{(j)} - \beta| = \frac{|\bar{\varepsilon}_j^Z|}{|\hat{\mu}_{ZX}^{(j)}|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}[|\hat{\beta}^{(j)} - \beta| > t] \leq \mathbb{P}[A_j(D)^c] + \mathbb{P}[B_j(D, t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). The two failure events are the block-level analogues of those in the proof of Theorem 3.3, with $\hat{\mu}_{ZX}$ replaced by the block mean $\hat{\mu}_{ZX}^{(j)}$, $\hat{\mu}_{Z\varepsilon}$ by $\bar{\varepsilon}_j^Z$, and the full sample size n by the block size m . The same reverse-triangle-inequality and Chebyshev arguments therefore give

$$\mathbb{P}[A_j^c] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}, \quad \mathbb{P}[B_j^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2}.$$

Step 3 (Allocate the error budget and solve for t). Unlike in the standard IV proof, the target here is not δ but the per-block failure probability $1/4$: this is the level at which the counting argument of Steps 4–5 boosts the median to the desired confidence. We require $\mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq 1/4$ and allocate half the budget to each event.

Budget for A_j^c . Setting $\mathbb{P}[A_j^c] \leq 1/8$ requires

$$\frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2} \leq \frac{1}{8} \iff m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2},$$

which is exactly condition (17).

Setting $\mathbb{P}[B_j^c] \leq 1/8$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2} \leq \frac{1}{8} \implies t \geq \frac{\sqrt{32}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}.$$

By the union bound, each block estimate satisfies

$$\mathbb{P}[|\hat{\beta}^{(j)} - \beta| > t] \leq \mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Step 4 (Counting argument). Define $\zeta_j = \mathbf{1}\{|\hat{\beta}^{(j)} - \beta| > t\}$. By Step 3, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint, the $(\zeta_j)_{j=1}^k$ are independent. If $|\hat{\beta}_{\text{MR}} - \beta| > t$, the median of the block estimates lies outside $(\beta - t, \beta + t)$, which requires at least $k/2$ blocks to satisfy $|\hat{\beta}^{(j)} - \beta| > t$. Therefore,

$$\mathbb{P}[|\hat{\beta}_{\text{MR}} - \beta| > t] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right].$$

Step 5 (Hoeffding). By Hoeffding's inequality, with overshoot $k/2 - k/4 = k/4$,

$$\mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ gives $e^{-k/8} \leq \delta$, completing the proof. $\square$

Remark 5.3 (Logarithmic dependence on δ). *Through $k = \lceil 8 \ln(1/\delta) \rceil$, the bound (18) depends on $1/\delta$ only logarithmically. This is precisely the improvement over the standard IV estimator anticipated in Remark 3.4: the median-of-block-estimates construction replaces the polynomial factor $1/\sqrt{\delta}$ of (8) by $\sqrt{\ln(1/\delta)}$.*

Remark 5.4 (The symmetric threshold is not optimal). *As in the standard IV estimator (Remark 3.5), the choice $D = |\mu_{ZX}|/2$ was made for simplicity rather than optimality. Section 5.1 treats D as a free parameter.*

5.1 Optimised Threshold

The proof of Theorem 5.2 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

Theorem 5.5 (MoR concentration, optimised threshold). *Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(1/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. Define*

$$\rho_{\text{MR}} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

If $\rho_{\text{MR}} < 1$ (i.e., $m > 4\sigma_{ZX}^2/\mu_{ZX}^2$), then the choice $D^ = (1 - \rho_{\text{MR}}^{1/3})|\mu_{ZX}|$ is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{MR}} - \beta\right| > \frac{2\sigma_{Z\epsilon}}{|\mu_{ZX}|\sqrt{m}(1 - \rho_{\text{MR}}^{1/3})^{3/2}}\right] \leq \delta. \quad (19)$$

Proof. Steps 1 and 2 are identical to those of Theorem 5.2 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A_j(D)^c] \leq \frac{\sigma_{ZX}^2/m}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B_j(D, t)^c] \leq \frac{\sigma_{Z\epsilon}^2/m}{t^2 D^2}.$$

Requiring their sum to be at most $\frac{1}{4}$ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{m(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\epsilon}^2}{mt^2 D^2} \leq \frac{1}{4}. \quad (20)$$

Step 3 (Optimise D , then solve for t). Writing $D = \eta|\mu_{ZX}|$ with $\eta \in (0, 1)$ and multiplying (20) through by 4, the constraint becomes $\rho_{\text{MR}}/(1 - \eta)^2 + 4\sigma_{Z\epsilon}^2/(m\mu_{ZX}^2\eta^2 t^2) \leq 1$. This is exactly the constraint optimised in the proof of Theorem 3.6, with ρ_{IV} replaced by ρ_{MR} and the

prefactor $\sigma_{Z\varepsilon}^2/(\delta n \mu_{ZX}^2)$ by $4\sigma_{Z\varepsilon}^2/(m \mu_{ZX}^2)$. The same optimisation therefore yields $\eta^* = 1 - \rho_{\text{MR}}^{1/3}$, feasible if and only if $\rho_{\text{MR}} < 1$, and

$$t^* = \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{m} (1 - \rho_{\text{MR}}^{1/3})^{3/2}}.$$

At this threshold each block satisfies $\mathbb{P}[\left|\hat{\beta}^{(j)} - \beta\right| > t^*] \leq 1/4$, and the counting argument and Hoeffding's inequality of Steps 4–5 in the proof of Theorem 5.2 then give $\mathbb{P}[\left|\hat{\beta}_{\text{MR}} - \beta\right| > t^*] \leq \delta$. $\square$

5.2 Cantelli Improvement

As in the standard IV estimator (Section 3.2), the denominator failure event $A_j(D)^c$ is one-sided: only a downward deviation of $\hat{\mu}_{ZX}^{(j)}$ from μ_{ZX} makes the denominator too small, while an upward deviation is harmless. Replacing the two-sided Chebyshev bound on this event by Cantelli's inequality (Lemma 3.8) sharpens the instrument strength condition.

Theorem 5.6 (MoR concentration, Cantelli improvement). *Assume (A1)–(A3), $\mu_{ZX} > 0$, and let $\delta \in (0, 1)$, $k = \lceil 8 \ln(1/\delta) \rceil$, $m = \lfloor n/k \rfloor$. Write $\gamma_{\text{MR}} = \sigma_{ZX}^2/(m \mu_{ZX}^2)$ and define*

$$h(\eta) = \frac{\eta^2((1 - \eta)^2 - 3\gamma_{\text{MR}})}{\gamma_{\text{MR}} + (1 - \eta)^2}, \quad \eta \in (0, 1). \quad (21)$$

If

$$m > \frac{3\sigma_{ZX}^2}{\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{\text{MR}} < \frac{1}{3}, \quad (22)$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{MR}} - \beta\right| > \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{mh(\eta_{\text{MR}}^*)}}\right] \leq \delta, \quad (23)$$

where η_{MR}^* is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta \mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). Since $\mu_{ZX} > 0$, we drop the absolute value on the denominator and define $A_j(D) = \{\hat{\mu}_{ZX}^{(j)} \geq D\}$ and $B_j(D, t) = \{|\bar{\varepsilon}_j^Z| \leq tD\}$. As in Theorem 5.2, $|\hat{\beta}^{(j)} - \beta| \leq t$ on $A_j(D) \cap B_j(D, t)$.

Step 2 (Cantelli on A_j^c , Chebyshev on B_j^c). The event $A_j(D)^c$ is a one-sided downward deviation, so Lemma 3.8 applies with $W = \hat{\mu}_{ZX}^{(j)}$, $\sigma^2 = \sigma_{ZX}^2/m$, and $\tau = \mu_{ZX} - D$. The event $B_j(D, t)^c = \{|\bar{\varepsilon}_j^Z| > tD\}$ is two-sided, so Chebyshev remains appropriate. As in the proof of Theorem 3.9, these give

$$\mathbb{P}[A_j(D)^c] \leq \frac{\sigma_{ZX}^2/m}{\sigma_{ZX}^2/m + (\mu_{ZX} - D)^2}, \quad \mathbb{P}[B_j(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2 D^2}.$$

Step 3 (Constraint and optimisation). Setting the per-block failure probability to at most

1/4 and substituting $D = \eta\mu_{ZX}$, $\gamma_{\text{MR}} = \sigma_{Z\epsilon}^2/(m\mu_{ZX}^2)$:

$$\frac{\gamma_{\text{MR}}}{\gamma_{\text{MR}} + (1 - \eta)^2} + \frac{\sigma_{Z\epsilon}^2}{mt^2\eta^2\mu_{ZX}^2} \leq \frac{1}{4}.$$

Isolating the second term and solving for t^2 :

$$t^2 \geq \frac{4\sigma_{Z\epsilon}^2}{m\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\frac{1}{4}(\gamma_{\text{MR}} + (1 - \eta)^2) > \gamma_{\text{MR}}$, i.e., $(1 - \eta)^2 > 3\gamma_{\text{MR}}$. The feasible region is non-empty if and only if $\gamma_{\text{MR}} < \frac{1}{3}$, which is condition (22). Minimising t over the feasible region by maximising h and applying Hoeffding and the union bound gives (23). $\square$

Remark 5.7 (No tractable form for η_{MR}^*). *Exactly as for the standard IV estimator (Remark 3.11), the first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no tractable closed-form solution, in contrast to the clean form $\eta^* = 1 - \rho_{\text{MR}}^{1/3}$ of the optimised Chebyshev version (Theorem 5.5). The bound (23) must therefore be evaluated numerically.*

6 Randomised Partitions and Aggregation

Every estimator of Sections 4–5 and every test of Section 7 begins by cutting the sample into k blocks, and the proofs require only that the blocks be disjoint, of equal size, and fixed in advance of the data. In an application, samples are commonly ordered, whether by date, region, or cohort, so blocks cut from the file as it stands would differ for reasons unrelated to sampling noise. Permuting the rows uniformly at random before cutting removes this dependence on the file order, at the cost of making the estimate a function of the seed. We show that the guarantees survive the randomisation, that they say little about a fixed sample under a varying seed, and that aggregating over permutations removes the seed dependence for a factor two in δ .

Definition 6.1 (Randomised and aggregated estimators). *Let $S = (Y_i, X_i, Z_i)_{i=1}^n$ be the sample and $S^{(\pi)} = (Y_{\pi(i)}, X_{\pi(i)}, Z_{\pi(i)})_{i=1}^n$ its reordering by a permutation π of $\{1, \dots, n\}$. Let $\hat{\beta}(S^{(\pi)})$ be any of the block-based estimators of Sections 4–5 computed on $S^{(\pi)}$, the blocks being the consecutive runs of $m = \lfloor n/k \rfloor$ positions, and let $\pi_1, \dots, \pi_B$ be drawn independently and uniformly at random, independently of S . The randomised estimator is $\hat{\beta}(S^{(\pi_1)})$ and the aggregated estimator is $\tilde{\beta}_B = \text{med}(\hat{\beta}(S^{(\pi_1)}), \dots, \hat{\beta}(S^{(\pi_B)}))$. Different aggregation methods can be used; we utilise the median. Write $\tilde{\beta}_\infty$ for the median of the $n!$ values $\hat{\beta}(S^{(p)})$, one for each permutation p ; the idealised aggregate obtained when every permutation is used.*

Proposition 6.2 (Transfer to the randomised estimator). *Assume (A1)–(A3) and let $\hat{\beta}$ be the Ratio-of-Medians or Median-of-Ratios estimator, with k , m and the instrument strength condition of Theorem 4.1 or Theorem 5.2 in force, and let $t = t(n, k, \delta)$ be the deviation appearing there, so that $\mathbb{P}_S \left[\left| \hat{\beta}(S) - \beta \right| > t \right] \leq \delta$. Then*

$$\mathbb{P}_{S, \pi} \left[\left| \hat{\beta}(S^{(\pi)}) - \beta \right| > t \right] \leq \delta. \quad (24)$$

Proof. Fix a permutation p . Since S is i.i.d. it is exchangeable, so $S^{(p)}$ has the same distribution as S , and the hypothesis applied to that common distribution gives

$$\mathbb{P}_S \left[\left| \hat{\beta}(S^{(p)}) - \beta \right| > t \right] \leq \delta. \quad (25)$$

As π is uniform on the $n!$ permutations and independent of S , conditioning on its value gives $\mathbb{P}_{S,\pi} \left[\left| \hat{\beta}(S^{(\pi)}) - \beta \right| > t \right] = \frac{1}{n!} \sum_p \mathbb{P}_S \left[\left| \hat{\beta}(S^{(p)}) - \beta \right| > t \right] \leq \delta$. $\square$

Randomising the blocking therefore costs nothing; the randomised estimator inherits the guarantee of a fixed partition, and the same argument applies to the size bound of Theorem 7.1. What the argument needs is that π be drawn independently of the data. A seed chosen after looking at S , such as the one giving the most favourable estimate, changes the distribution of the reordered sample, and (24) may then fail. Note also that (24) averages over the sample and the seed together. An applied researcher is in a different position, holding the sample fixed and varying only the seed.

Proposition 6.3 (Conditional failure fraction). *In the setting of Proposition 6.2, let*

$$W(S) = \frac{1}{n!} \sum_p \mathbf{1} \left\{ \left| \hat{\beta}(S^{(p)}) - \beta \right| > t \right\} \quad (26)$$

be the fraction of permutations that fail on the sample S . Then $\mathbb{E}_S[W(S)] \leq \delta$ and $\mathbb{P}_S[W(S) \geq \lambda] \leq \delta/\lambda$ for every $\lambda \in (0, 1)$.

Proof. Taking expectations over S in (26) and exchanging the finite sum with the expectation, $\mathbb{E}_S[W(S)] = \frac{1}{n!} \sum_p \mathbb{P}_S \left[\left| \hat{\beta}(S^{(p)}) - \beta \right| > t \right] \leq \delta$, each term being bounded by (25). Since $W(S) \geq 0$, Markov's inequality gives $\mathbb{P}_S[W(S) \geq \lambda] \leq \mathbb{E}_S[W(S)]/\lambda \leq \delta/\lambda$. $\square$

At $\lambda = 1/2$ this says only that, outside a 2δ fraction of samples, fewer than half of the permutations fail. It says nothing about how far apart the resulting estimates are, which is instead measured empirically in Section 8.3.8.

Lemma 6.4 (Majority in an interval). *Let $x_1, \dots, x_N \in \mathbb{R}$ and let $I \subseteq \mathbb{R}$ be an interval containing more than $N/2$ of the x_j . Then every median of $x_1, \dots, x_N$ lies in I .*

Proof. Let a be the number of x_j lying below I and c the number lying above, so that $a+c < N/2$. Sorting the values as $x_{(1)} \leq \dots \leq x_{(N)}$, those in I occupy the positions $a+1, \dots, N-c$. From $a < N/2$ we get $a+1 \leq \lceil N/2 \rceil$, and from $c < N/2$ we get $N-c \geq \lfloor N/2 \rfloor + 1$, so the two central values $x_{(\lceil N/2 \rceil)}$ and $x_{(\lfloor N/2 \rfloor + 1)}$ both lie in I . Every median lies between them, and I is an interval, so it lies in I as well. $\square$

Theorem 6.5 (Concentration of the aggregated estimator). *In the setting of Proposition 6.2, the aggregated estimator of Definition 6.1 satisfies, for every $B \geq 1$ and every $\lambda \in (0, 1/2)$,*

$$\mathbb{P}_S \left[\left| \tilde{\beta}_\infty - \beta \right| > t \right] \leq 2\delta, \quad \mathbb{P} \left[\left| \tilde{\beta}_B - \beta \right| > t \right] \leq \frac{\delta}{\lambda} + e^{-2B(1/2-\lambda)^2}. \quad (27)$$

Proof. Step 1 (All permutations). Write $I = [\beta - t, \beta + t]$, so that a permutation p fails precisely when $\hat{\beta}(S^{(p)}) \notin I$. If $W(S) < 1/2$ then more than half of the $n!$ values $\hat{\beta}(S^{(p)})$ lie in I ,

so $\tilde{\beta}_\infty \in I$ by Lemma 6.4. Hence $\mathbb{P}_S[|\tilde{\beta}_\infty - \beta| > t] \leq \mathbb{P}_S[W(S) \geq 1/2] \leq 2\delta$ by Proposition 6.3 at $\lambda = 1/2$.

Step 2 (Finitely many permutations). Fix the sample S and set $\zeta_b = \mathbf{1}\{\hat{\beta}(S^{(\pi_b)}) \notin I\}$. As each π_b is uniform and independent of S , the probability that it fails is exactly $W(S)$, so $\zeta_1, \dots, \zeta_B$ are i.i.d. Bernoulli with mean $W(S)$. If fewer than $B/2$ of them equal one, more than half of the values $\hat{\beta}(S^{(\pi_b)})$ lie in I , and Lemma 6.4 gives $\tilde{\beta}_B \in I$. Whenever $W(S) \leq \lambda$, the aggregate can therefore fail only if $\sum_b \zeta_b$ exceeds its mean by at least $B(1/2 - \lambda)$, so Hoeffding's inequality gives

$$\mathbb{P}\left[|\tilde{\beta}_B - \beta| > t \mid S\right] \leq \mathbb{P}\left[\sum_b \zeta_b \geq B/2 \mid S\right] \leq e^{-2B(1/2-\lambda)^2}.$$

Averaging over S now gives the second bound: the display applies to every sample with $W(S) \leq \lambda$, and the remaining samples occur with probability at most δ/λ by Proposition 6.3. $\square$

Remark 6.6 (The price, in blocks). *The two bounds in (27) price the aggregation differently. For the idealised aggregate $\tilde{\beta}_\infty$, running the underlying estimator at $\delta/2$ recovers the original bound at level δ ; for the Median-of-Ratios estimator this replaces $k = \lceil 8 \ln(1/\delta) \rceil$ by $k = \lceil 8 \ln(2/\delta) \rceil$, the block count the Ratio-of-Medians estimator already pays for its union bound in Theorem 4.1. For finite B the leading term is δ/λ rather than 2δ , so a choice of λ must be fixed: at $\lambda = 1/3$, running the underlying estimator at $\delta/3$, i.e. $k = \lceil 8 \ln(3/\delta) \rceil$, gives $\delta/\lambda + e^{-2B(1/2-\lambda)^2} \leq \delta + e^{-B/18}$, and the exponential term is negligible, contributing at most $2 \cdot 10^{-5}$ at $B = 200$. In exchange, the estimate no longer depends on a seed.*

Remark 6.7 (Why the median and not the mean). *Theorem 6.5 uses only that a majority of permutations succeed; the failing minority may be arbitrarily bad without moving the median outside I . Averaging does not have the same guarantee, since one extreme value moves the mean by an arbitrary amount. Since the estimators are ratios, and a partition placing an unusually small value in a denominator produces an arbitrarily large estimate, which a mean would pass on.*

Remark 6.8 (The permutation distribution is not a sampling distribution). *With S held fixed the draws $\hat{\beta}(S^{(\pi_b)})$ are i.i.d., so their spread shrinks as B grows; it measures the computational randomness of the blocking, not uncertainty about β . An interval formed from their quantiles describes only how the estimate moves as the seed varies on the sample in hand, and must not be reported as one for β .*

7 Inference

7.1 Test Inversion

Fix a null hypothesis $H_0: \beta = \beta_0$ and define

$$W_i(\beta_0) := Z_i(Y_i - \beta_0 X_i), \quad i = 1, \dots, n. \quad (28)$$

Under H_0 we have $W_i(\beta_0) = Z_i \varepsilon_i$, so by (A1) $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i \varepsilon_i] = 0$ and by (A3) $\text{Var}(W_i(\beta_0)) = \sigma_{Z\varepsilon}^2 < \infty$.

Dufour (1997) explains why test inversion is necessary under weak instruments. Let $\mathcal{P}$ denote the family of joint distributions of (Y_i, X_i, Z_i) compatible with (A1) and (A3), note that (A2) is deliberately not imposed, and write $\mu_{ZX}(P) := \mathbb{E}_P[Z_i X_i]$ for $P \in \mathcal{P}$. On the identification failure set

$$\mathcal{P}_0 := \{P \in \mathcal{P} : \mu_{ZX}(P) = 0\}$$

the parameter β is unidentified; exogeneity then forces $\mu_{ZY}(P) = 0$ as well, so $\mathbb{E}_P[Z_i(Y_i - bX_i)] = 0$ for every $b \in \mathbb{R}$. Since $\mathcal{P}_0$ is non-empty, no confidence set procedure that is valid over all of $\mathcal{P}$ can be bounded with probability one. A Wald-type interval built around a point estimate is therefore not generally valid, whereas inverting a test of $H_0: \beta = \beta_0$ over a grid of null values remains so.

7.2 The Median-of-Means Anderson–Rubin Test

We partition the sample into k blocks as in the preceding sections and form, within each block B_j , the block mean of the moment (28). Because $W_i(\beta_0) = Z_i Y_i - \beta_0 Z_i X_i$ is linear in β_0 , the block mean is an *affine* function of β_0 :

$$\bar{W}_j(\beta_0) := \frac{1}{m} \sum_{i \in B_j} W_i(\beta_0) = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}, \quad j = 1, \dots, k. \quad (29)$$

The median-of-means AR statistic is the coordinate-wise median of these block means,

$$\widetilde{W}(\beta_0) := \text{med}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)). \quad (30)$$

Algorithm 4: Median-of-Means Anderson–Rubin Test

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k , null value β_0 , threshold $\tau_n(\delta)$

Output: Decision: reject or fail to reject $H_0: \beta = \beta_0$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
Form $\bar{W}_j(\beta_0) = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}$;

$\widetilde{W}(\beta_0) \leftarrow \text{med}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0))$;

return reject if $|\widetilde{W}(\beta_0)| > \tau_n(\delta)$, else fail to reject;

Theorem 7.1 (Size of the MoM-AR test). *Assume (A1)–(A3). Fix $\delta \in (0, 1)$, set $k = \lceil 8 \ln(1/\delta) \rceil$ and $m = \lfloor n/k \rfloor$, and define*

$$\tau_n(\delta) := \sigma_{Z\varepsilon} \sqrt{\frac{32 \ln(1/\delta)}{n}}. \quad (31)$$

Then under $H_0: \beta = \beta_0$,

$$\mathbb{P}_{H_0} \left[|\widetilde{W}(\beta_0)| > \tau_n(\delta) \right] \leq \delta. \quad (32)$$

Consequently the test of Algorithm 4 has size at most δ .

Proof. The proof reduces the statistic to the scalar median-of-means estimator of Section 2.3 and applies its concentration bound.

Step 1 (Moments of the block moment under H_0). Under $H_0: \beta = \beta_0$ we have $W_i(\beta_0) = Z_i \varepsilon_i$, so the $(W_i(\beta_0))_{i=1}^n$ are i.i.d. with $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i \varepsilon_i] = 0$ by (A1) and $\text{Var}(W_i(\beta_0)) = \sigma_{Z\varepsilon}^2 < \infty$ by (A3).

Step 2 (Reduction to the scalar MoM estimator). By (29) and (30), $\widetilde{W}(\beta_0)$ is the coordinate-wise median of the k block means of $(W_i(\beta_0))_{i=1}^n$; that is, it is exactly the scalar median-of-means estimator of Algorithm 1 applied to this sequence, estimating its mean $\mu = 0$. Theorem 2.4, taken with $\sigma^2 = \sigma_{Z\varepsilon}^2$ and $k = \lceil 8 \ln(1/\delta) \rceil$, therefore gives

$$\mathbb{P}_{H_0} \left[\left| \widetilde{W}(\beta_0) \right| > \sigma_{Z\varepsilon} \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta.$$

The deviation on the left is precisely $\tau_n(\delta)$ of (31), so this is (32). $\square$

7.3 The MoM-AR Confidence Set

Inverting the test of Theorem 7.1 produces a confidence set.

Definition 7.2 (MoM-AR confidence set). *For $\delta \in (0, 1)$ and $\tau_n(\delta)$ as in (31), the MoM-AR confidence set is*

$$CS_{1-\delta} := \{ \beta_0 \in \mathbb{R} : \left| \widetilde{W}(\beta_0) \right| \leq \tau_n(\delta) \}. \quad (33)$$

Theorem 7.3 (Coverage). *Assume (A1)–(A3) and let $k = \lceil 8 \ln(1/\delta) \rceil$. Then*

$$\mathbb{P}_\beta [\beta \in CS_{1-\delta}] \geq 1 - \delta.$$

Proof. The result is the test-inversion dual of the size bound. By Definition 7.2, the true parameter lies in the confidence set exactly when the statistic evaluated at the truth stays below the threshold:

$$\beta \in CS_{1-\delta} \iff \left| \widetilde{W}(\beta) \right| \leq \tau_n(\delta).$$

Evaluating the moment at the true parameter gives $W_i(\beta) = Z_i \varepsilon_i$, which has mean zero by (A1). Applying the size bound of Theorem 7.1 at the true null $\beta_0 = \beta$ therefore yields $\mathbb{P}_\beta \left[\left| \widetilde{W}(\beta) \right| > \tau_n(\delta) \right] \leq \delta$. Taking complements,

$$\mathbb{P}_\beta [\beta \in CS_{1-\delta}] = \mathbb{P}_\beta \left[\left| \widetilde{W}(\beta) \right| \leq \tau_n(\delta) \right] \geq 1 - \delta. \quad \square$$

7.4 Geometry of the Confidence Set

Theorem 7.4 (Piecewise-affine structure). *The map $\beta_0 \mapsto \widetilde{W}(\beta_0)$ is continuous and piecewise affine. There exist finitely many breakpoints $\beta^{(1)} < \beta^{(2)} < \dots < \beta^{(M)}$, with $M \leq \binom{k}{2}$, such that on each open interval of $\mathbb{R} \setminus \{\beta^{(1)}, \dots, \beta^{(M)}\}$ the function $\widetilde{W}$ coincides with a single block mean $\bar{W}_j$ and is therefore affine.*

Proof. The median of k numbers is the value occupying rank $r := \lceil k/2 \rceil$ once the numbers are sorted. We therefore study how the sorted order of $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$ varies with β_0 .

Cite
confid-
ence set

Step 1 (Crossings are finite). For a pair $i \neq j$, the equation $\bar{W}_i(\beta_0) = \bar{W}_j(\beta_0)$ reads $\hat{\mu}_{ZY}^{(i)} - \beta_0 \hat{\mu}_{ZX}^{(i)} = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}$, i.e. $(\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\beta_0 = \hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)}$. If $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$ this has a unique solution; if the slopes are equal the lines are parallel and never cross (or coincide, in which case the pair may be merged). Hence the crossing set $\mathcal{C} := \{\beta_0 : \bar{W}_i(\beta_0) = \bar{W}_j(\beta_0) \text{ for some } i \neq j\}$ satisfies $|\mathcal{C}| \leq \binom{k}{2}$. Enumerate it as $\beta^{(1)} < \dots < \beta^{(M)}$ and set $\beta^{(0)} := -\infty$, $\beta^{(M+1)} := +\infty$.

Step 2 (The order is constant between crossings). Fix an interval $I_\ell := (\beta^{(\ell)}, \beta^{(\ell+1)})$. For any pair $i \neq j$ the difference $D_{ij}(\beta_0) := \bar{W}_i(\beta_0) - \bar{W}_j(\beta_0)$ is affine, hence continuous, and vanishes only on $\mathcal{C}$. Since $I_\ell \cap \mathcal{C} = \emptyset$, D_{ij} has constant sign on I_ℓ . As this holds for every pair, the total ordering of $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$ is constant on I_ℓ ; let π_ℓ be the permutation realising it, so that $\bar{W}_{\pi_\ell(1)}(\beta_0) \leq \dots \leq \bar{W}_{\pi_\ell(k)}(\beta_0)$ throughout I_ℓ .

Step 3 (The median is one block mean on each interval). By Step 2 the rank- r value on I_ℓ is always supplied by the same index, so $\widetilde{W}(\beta_0) = \bar{W}_{\pi_\ell(r)}(\beta_0)$ for all $\beta_0 \in I_\ell$; this is affine. With at most $M + 1 \leq \binom{k}{2} + 1$ intervals, $\widetilde{W}$ is piecewise affine.

Step 4 (Continuity). Let $i = \pi_{\ell-1}(r)$ and $j = \pi_\ell(r)$ be the active median indices on either side of a breakpoint $\beta^{(\ell)}$. If $i = j$ there is nothing to prove. If $i \neq j$, the rank- r slot changes occupant across $\beta^{(\ell)}$, which can occur only if lines i and j exchange ranks there; by continuity of D_{ij} this requires $\bar{W}_i(\beta^{(\ell)}) = \bar{W}_j(\beta^{(\ell)})$. Hence the left and right limits of $\widetilde{W}$ at $\beta^{(\ell)}$ coincide with this common value, and $\widetilde{W}$ is continuous. (If three or more lines meet at $\beta^{(\ell)}$, any reshuffling is confined to the indices sharing the common value there, and the same conclusion holds.) $\square$

Corollary 7.5 (Candidate boundary points). *Every boundary point of $CS_{1-\delta}$ lies in the set*

$$\mathcal{B}_{\text{cand}} := \bigcup_{j: \hat{\mu}_{ZX}^{(j)} \neq 0} \left\{ \frac{\hat{\mu}_{ZY}^{(j)} - \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}}, \frac{\hat{\mu}_{ZY}^{(j)} + \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}} \right\}, \quad (34)$$

a set of at most $2k$ explicitly computable points.

Proof. A boundary point β_0 of $CS_{1-\delta}$ lies on the boundary of the band, so $|\widetilde{W}(\beta_0)| = \tau_n(\delta)$, i.e. $\widetilde{W}(\beta_0) = +\tau_n(\delta)$ or $\widetilde{W}(\beta_0) = -\tau_n(\delta)$. By Theorem 7.4, β_0 lies in the closure of an interval on which $\widetilde{W}$ coincides with a single block mean $\bar{W}_j$, so it solves $\bar{W}_j(\beta_0) = \pm\tau_n(\delta)$ for some j . Two cases arise, according to the slope $-\hat{\mu}_{ZX}^{(j)}$ of that piece.

If $\hat{\mu}_{ZX}^{(j)} \neq 0$, then $\bar{W}_j$ is strictly monotone and $\hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)} = \pm\tau_n(\delta)$ has the unique solutions

$$\beta_0 = \frac{\hat{\mu}_{ZY}^{(j)} \mp \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}},$$

namely the two values contributed by block j in (34). If $\hat{\mu}_{ZX}^{(j)} = 0$, then $\bar{W}_j \equiv \hat{\mu}_{ZY}^{(j)}$ is constant and equals $\pm\tau_n(\delta)$ only in the non-generic event $\hat{\mu}_{ZY}^{(j)} = \pm\tau_n(\delta)$; such a block contributes no isolated boundary point. Ranging over the k blocks, each of which contributes at most two candidates, gives $|\mathcal{B}_{\text{cand}}| \leq 2k$. $\square$

Corollary 7.6 (Finite union of intervals). *$CS_{1-\delta}$ is a finite union of at most $k + 1$ closed intervals, each possibly a single point or unbounded.*

Proof. By Theorem 7.4, $|\widetilde{W}|$ is continuous and piecewise affine. The set $CS_{1-\delta} = \{\beta_0 : |\widetilde{W}(\beta_0)| \leq \tau_n(\delta)\}$ is the preimage of the closed interval $[-\tau_n(\delta), \tau_n(\delta)]$ under this continuous function, hence closed; being a sublevel set of a piecewise-affine function with finitely many pieces, it is a finite union of closed intervals.

To count the components, note that the endpoints of these intervals are boundary points of $CS_{1-\delta}$, and by Corollary 7.5 there are at most $2k$ such points. Ordering them partitions $\mathbb{R}$ into at most $2k + 1$ open intervals, on each of which $|\widetilde{W}| - \tau_n(\delta)$ has constant sign and so lies either entirely inside or entirely outside the band. Since the in-band and out-of-band intervals alternate along the line, at most $\lceil (2k + 1)/2 \rceil = k + 1$ of them lie in the band. Merging any in-band intervals that meet at a shared boundary point can only reduce this count, so $CS_{1-\delta}$ has at most $k + 1$ components. $\square$

7.5 Monotonicity and Single-Interval Confidence Sets

The slope of $\widetilde{W}$ on any affine piece is $-\hat{\mu}_{ZX}^{(j)}$ for the active index j . When all block slopes share a sign, $\widetilde{W}$ is monotonic and the confidence set is a single interval.

Proposition 7.7 (Deterministic single-interval condition). *If the block sums $\hat{\mu}_{ZX}^{(1)}, \dots, \hat{\mu}_{ZX}^{(k)}$ all share the same sign, then $\widetilde{W}$ is monotonic on $\mathbb{R}$ and $CS_{1-\delta}$ is a single (possibly empty or unbounded) interval.*

Proof. By Theorem 7.4 the slope of $\widetilde{W}$ on each affine piece equals $-\hat{\mu}_{ZX}^{(j)}$ for the active index j . If all $\hat{\mu}_{ZX}^{(j)} > 0$ then every such slope is negative, so the continuous piecewise-affine function $\widetilde{W}$ is non-increasing on $\mathbb{R}$; if all $\hat{\mu}_{ZX}^{(j)} < 0$ it is non-decreasing. In either case $\widetilde{W}$ is monotone, so each of the level sets $\{\widetilde{W} \leq \tau_n(\delta)\}$ and $\{\widetilde{W} \geq -\tau_n(\delta)\}$ is a half-line (a ray). The confidence set $CS_{1-\delta}$ is their intersection, hence a single interval, which may be empty or unbounded. $\square$

Proposition 7.8 (Single-interval condition, Chebyshev). *Assume (A1)–(A3) with $\mu_{ZX} \neq 0$. If*

$$m \geq \frac{k \sigma_{ZX}^2}{\delta \mu_{ZX}^2}, \quad (35)$$

then with probability at least $1 - \delta$ all block sums $\hat{\mu}_{ZX}^{(j)}$ share the sign of μ_{ZX} ; consequently $\widetilde{W}$ is monotone and $CS_{1-\delta}$ is a single interval.

Proof. Assume without loss of generality that $\mu_{ZX} > 0$; the case $\mu_{ZX} < 0$ is symmetric. By Proposition 7.7 it suffices to show that all block sums are positive with probability at least $1 - \delta$, since sign-consistency then forces $\widetilde{W}$ to be monotone. Sign-consistency fails on the event

$$\mathcal{F} = \bigcup_{j=1}^k \{\hat{\mu}_{ZX}^{(j)} \leq 0\}.$$

Step 1 (Chebyshev on each block). A non-positive block sum is a large downward deviation from $\mu_{ZX} > 0$: if $\hat{\mu}_{ZX}^{(j)} \leq 0$ then $|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}| = \mu_{ZX} - \hat{\mu}_{ZX}^{(j)} \geq \mu_{ZX}$, so $\{\hat{\mu}_{ZX}^{(j)} \leq 0\} \subseteq \{|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}|\geq \mu_{ZX}\}$. Since $\text{Var}(\hat{\mu}_{ZX}^{(j)}) = \sigma_{ZX}^2/m$, Chebyshev's inequality applied with

threshold μ_{ZX} gives

$$\mathbb{P}[\hat{\mu}_{ZX}^{(j)} \leq 0] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2} = \frac{\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

Step 2 (Union bound over blocks). Summing over the k blocks,

$$\mathbb{P}[\mathcal{F}] \leq \sum_{j=1}^k \mathbb{P}[\hat{\mu}_{ZX}^{(j)} \leq 0] \leq \frac{k\sigma_{ZX}^2}{m\mu_{ZX}^2},$$

which is at most δ precisely when $m \geq k\sigma_{ZX}^2/(\delta\mu_{ZX}^2)$, i.e. condition (35). On the complement $\mathcal{F}^c$, all block sums share the sign of μ_{ZX} , and Proposition 7.7 makes $CS_{1-\delta}$ a single interval. $\square$

Remark 7.9 (Cantelli refinement). *Replacing Chebyshev's inequality by the one-sided Cantelli bound (Lemma 3.8) in Step 1 sharpens the per-block probability, and hence the constant in condition (35). As for the standard IV estimator (Remark 3.10), the resulting improvement is minor, so we retain the Chebyshev form.*

7.6 Computation

The confidence set can be computed exactly and in closed form. The idea is to enumerate the crossing points of the block-mean lines $\bar{W}_1, \dots, \bar{W}_k$ to obtain the affine pieces of $\widetilde{W}$ (Theorem 7.4), identify the active median line on each piece, and retain the sub-intervals on which this line lies within the band $[-\tau_n(\delta), \tau_n(\delta)]$. Algorithm 5 collects these steps.

Algorithm 5: MoM-AR Confidence Set

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k , threshold $\tau_n(\delta)$

Output: Confidence set $CS_{1-\delta}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;

$\mathcal{C} \leftarrow \emptyset$;

foreach pair $1 \leq i < j \leq k$ with $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$ **do**

$\mathcal{C} \leftarrow \mathcal{C} \cup \{(\hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)})/(\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\}$;

Sort $\mathcal{C}$ as $\beta^{(1)} < \dots < \beta^{(M)}$ and set $\beta^{(0)} = -\infty$, $\beta^{(M+1)} = +\infty$;

$CS_{1-\delta} \leftarrow \emptyset$;

for $\ell = 0, \dots, M$ **do**

 Choose any $\beta_\star \in (\beta^{(\ell)}, \beta^{(\ell+1)})$;

 Find the active median index a with $\bar{W}_a(\beta_\star) = \text{med}(\bar{W}_1(\beta_\star), \dots, \bar{W}_k(\beta_\star))$;

if $\hat{\mu}_{ZX}^{(a)} \neq 0$ **then**

$J \leftarrow$ closed interval with endpoints $(\hat{\mu}_{ZY}^{(a)} \pm \tau_n(\delta))/\hat{\mu}_{ZX}^{(a)}$;

else

$J \leftarrow \mathbb{R}$ if $|\hat{\mu}_{ZY}^{(a)}| \leq \tau_n(\delta)$, else $J \leftarrow \emptyset$;

$CS_{1-\delta} \leftarrow CS_{1-\delta} \cup ([\beta^{(\ell)}, \beta^{(\ell+1)}] \cap J)$;

return $CS_{1-\delta}$, merging intervals that share an endpoint;

On each interval $(\beta^{(\ell)}, \beta^{(\ell+1)})$ the statistic $\widetilde{W}$ coincides with a single affine piece $\bar{W}_a$, so

intersecting the band constraint $|\bar{W}_a(\beta_0)| \leq \tau_n(\delta)$ with that interval is exact. By Corollary 7.6 the returned set consists of at most $k + 1$ closed intervals, whose endpoints lie among the $2k$ candidates of Corollary 7.5.

Remark 7.10 (Feasibility and choice of variance estimator). *In practice, $\sigma_{Z\varepsilon}$ is unknown and has to be estimated with a consistent estimator. A feasible test would replace this by a consistent estimator $\hat{\sigma}_{Z\varepsilon}$. This estimator must be computed independently of the null value β_0 ; test inversion compares the varying statistic $\widetilde{W}(\beta_0)$ against a single band across a grid of β_0 values. Forming residuals at the tested value is inconsistent whenever $\beta_0 \neq \beta$ since $Y_i - \beta_0 X_i = \varepsilon_i + (\beta - \beta_0)X_i$,*

$$\frac{1}{n} \sum_{i=1}^n (Z_i(Y_i - \beta_0 X_i))^2 \xrightarrow{p} \sigma_{Z\varepsilon}^2 + 2(\beta - \beta_0) \mathbb{E}[Z^2 X \varepsilon] + (\beta - \beta_0)^2 \mathbb{E}[(ZX)^2],$$

which grows as β_0 departs from β . This would lead to the band widening with the signal it is trying to detect, leading to a loss of power and a collapse of the constant-threshold geometry of Section 7.4.

A β_0 -free estimate could be obtained instead from residuals at a preliminary consistent estimator. However, under the heavy tails permitted by (A3), $Z\varepsilon$ may have no finite fourth moment, in which case the estimator $\frac{1}{n} \sum_i (Z_i \hat{\varepsilon}_i)^2$ has infinite variance; the central limit theorem then no longer applies, so it converges slowly and behaves erratically. A robust scale estimator is therefore preferable.

The self-normalised variant discussed in Section 7.7 avoids the matter altogether, cancelling $\sigma_{Z\varepsilon}$ rather than estimating it.

7.7 Self-Normalised Anderson-Rubin Test

Definition 7.11 (Self-normalised statistic). *For a vector $v = (v_1, \dots, v_k) \in \mathbb{R}^k$, let $\text{MAD}(v) = \text{med}_j |v_j - \text{med}_l v_l|$ denote its median absolute deviation. With $\bar{W}_j$ and $\widetilde{W}$ as in (29)–(30), the self-normalised statistic at β_0 is*

$$T(\beta_0) = \frac{|\widetilde{W}(\beta_0)|}{\hat{s}(\beta_0)}, \quad \hat{s}(\beta_0) = \text{MAD}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)).$$

Given a critical value $c_{k,\delta}$, the test rejects $H_0: \beta = \beta_0$ when $T(\beta_0) > c_{k,\delta}$.

Proposition 7.12 (Scale invariance). *Under H_0 , the distribution of $T(\beta_0)$ does not depend on $\sigma_{Z\varepsilon}$. The same holds for any scale functional $\hat{s}$ that is homogeneous of degree one.*

Proof. Under H_0 , $\bar{W}_j(\beta_0) = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i = \sigma_{Z\varepsilon} G_j$, where $G_j = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i / \sigma_{Z\varepsilon}$. The summands $Z_i \varepsilon_i / \sigma_{Z\varepsilon}$ have mean 0 and variance 1 by (A1) and (A3), so the joint distribution $(G_1, \dots, G_k)$ are free of $\sigma_{Z\varepsilon}$. Since med and MAD are homogeneous of degree one,

$$T(\beta_0) = \frac{|\text{med}_j \sigma_{Z\varepsilon} G_j|}{\text{MAD}_j(\sigma_{Z\varepsilon} G_j)} = \frac{\sigma_{Z\varepsilon} |\text{med}_j G_j|}{\sigma_{Z\varepsilon} \text{MAD}_j(G_j)} = \frac{|\text{med}_j G_j|}{\text{MAD}_j(G_j)},$$

and a function of $(G_1, \dots, G_k)$ alone; its distribution therefore does not involve $\sigma_{Z\varepsilon}$. The argument uses only $\text{med}(\sigma v) = \sigma \text{med}(v)$ and $\hat{s}(\sigma v) = \sigma \hat{s}(v)$, so it holds for any degree-one homogeneous $\hat{s}$. $\square$

Proposition 7.13 (Asymptotic null distribution). *Assume (A1)–(A3), with k fixed and $m \rightarrow \infty$. Under H_0 ,*

$$T(\beta_0) \implies R_k := \frac{|\text{med}(\xi_1, \dots, \xi_k)|}{\text{MAD}(\xi_1, \dots, \xi_k)}, \quad \xi_1, \dots, \xi_k \stackrel{iid}{\sim} N(0, 1),$$

and the distribution of R_k depends only on k .

Proof. Fix a block. By (A1) and (A3) the standardised summands $Z_i \varepsilon_i / \sigma_{Z\varepsilon}$ are i.i.d. with mean 0 and variance 1, so the central limit theorem gives $\sqrt{m} G_j = \frac{1}{\sqrt{m}} \sum_{i \in B_j} Z_i \varepsilon_i / \sigma_{Z\varepsilon} \Rightarrow \xi_j \sim N(0, 1)$. The blocks use disjoint observations, so $(\sqrt{m} G_1, \dots, \sqrt{m} G_k) \Rightarrow (\xi_1, \dots, \xi_k)$ with independent coordinates. Writing $T(\beta_0) = |\text{med}_j \sqrt{m} G_j| / \text{MAD}_j(\sqrt{m} G_j)$, the factor $\sqrt{m}$ cancels between numerator and denominator; as med and MAD are continuous, the continuous mapping theorem yields $T(\beta_0) \Rightarrow R_k$. The limit is a fixed functional of k standard normals, so its distribution depends only on k . $\square$

The critical value $c_{k,\delta}$ is the $(1-\delta)$ quantile of R_k ; since R_k depends only on k , it is computed once by simulation and tabulated in Section 11.1

Corollary 7.14 (Confidence set). *Inverting the self-normalised test gives*

$$CS_{1-\delta} = \{\beta_0 \in \mathbb{R} : |\widetilde{W}(\beta_0)| \leq c_{k,\delta} \hat{s}(\beta_0)\},$$

a finite union of closed intervals.

Proof. By Theorem 7.4, $\widetilde{W}$ is continuous and piecewise affine; $\hat{s}(\beta_0) = \text{MAD}_j(\bar{W}_j(\beta_0))$ is a median of absolute values of affine functions, hence also continuous and piecewise affine. On any interval where both coincide with single affine pieces $a(\beta_0)$ and $b(\beta_0) \geq 0$, the defining condition $-c_{k,\delta} b(\beta_0) \leq a(\beta_0) \leq c_{k,\delta} b(\beta_0)$ is a pair of linear inequalities, hence an interval. As there are finitely many pieces, $CS_{1-\delta}$ is a finite union of closed intervals. $\square$

Remark 7.15 (An exactly distribution-free variant). *If $Z\varepsilon$ is symmetric about 0 under H_0 , the signs of the block means are independent fair coin flips, so $N_+(\beta_0) = \#\{j : \bar{W}_j(\beta_0) > 0\} \sim \text{Binomial}(k, \frac{1}{2})$ exactly, free of both $\sigma_{Z\varepsilon}$ and the tail shape. Rejecting when $N_+(\beta_0)$ is far from $k/2$ gives a finite-sample-exact, distribution-free test, at the cost of the symmetry assumption and of lower power, since it uses only the signs of the block means.*

7.8 Aggregated Inference

The tests of Section 7 partition the sample exactly as the estimators do, and the transfer argument of Proposition 6.2 applies to them unchanged. For any fixed permutation p the reordered sample $S^{(p)}$ has the same distribution as S , so the test of Algorithm 4 computed on $S^{(p)}$ has size at most δ by Theorem 7.1. What does not transfer is the verdict. Section 8.3.8 shows that, once the instrument weakens, whether the null is rejected can depend on the seed on a majority of datasets. We remove the dependence on the seed by taking a majority vote over permutations.

Definition 7.16 (Randomised and aggregated tests). *Fix a null value β_0 . For a permutation p , write $\widetilde{W}^{(p)}(\beta_0)$ for the statistic (30) computed on $S^{(p)}$ and $\varphi^{(p)}(\beta_0) = \mathbf{1}\{|\widetilde{W}^{(p)}(\beta_0)| > \tau_n(\delta)\}$ for the decision of Algorithm 4 on $S^{(p)}$, with 1 standing for rejection. With $\pi_1, \dots, \pi_B$ as in Definition 6.1, the randomised test is $\varphi^{(\pi_1)}(\beta_0)$ and the aggregated test rejects when more than half of the replicates do,*

$$\tilde{\varphi}_B(\beta_0) = \mathbf{1}\left\{\sum_{b=1}^B \varphi^{(\pi_b)}(\beta_0) > B/2\right\}. \quad (36)$$

Write $V(S, \beta_0) = \frac{1}{n!} \sum_p \varphi^{(p)}(\beta_0)$ for the fraction of all permutations that reject, and $\tilde{\varphi}_\infty(\beta_0) = \mathbf{1}\{V(S, \beta_0) > 1/2\}$ for the idealised test that consults every permutation.

Theorem 7.17 (Size of the aggregated test). *Assume (A1)–(A3), with $k = \lceil 8 \ln(1/\delta) \rceil$ and $\tau_n(\delta)$ as in (31). Then under $H_0: \beta = \beta_0$, for every $B \geq 1$ and every $\gamma \in (0, 1/2)$,*

$$\mathbb{P}_{H_0}[\tilde{\varphi}_\infty(\beta_0) = 1] \leq 2\delta, \quad \mathbb{P}_{H_0}[\tilde{\varphi}_B(\beta_0) = 1] \leq \frac{\delta}{\gamma} + e^{-2B(1/2-\gamma)^2}. \quad (37)$$

Proof. The argument is that of Theorem 6.5 with rejection in place of failure.

Step 1 (The rejecting fraction is small on average). Fix a permutation p . Under H_0 the reordered sample $S^{(p)}$ has the same distribution as S , so Theorem 7.1 applied to it gives $\mathbb{P}_{H_0}[\varphi^{(p)}(\beta_0) = 1] \leq \delta$. Taking expectations over S in the definition of V and exchanging the finite sum with the expectation, $\mathbb{E}_S[V(S, \beta_0)] = \frac{1}{n!} \sum_p \mathbb{P}_{H_0}[\varphi^{(p)}(\beta_0) = 1] \leq \delta$, and Markov's inequality gives $\mathbb{P}_S[V(S, \beta_0) \geq \gamma] \leq \delta/\gamma$ for every $\gamma \in (0, 1)$.

Step 2 (All permutations). The idealised test rejects exactly when $V(S, \beta_0) > 1/2$, so $\mathbb{P}_{H_0}[\tilde{\varphi}_\infty(\beta_0) = 1] \leq \mathbb{P}_S[V(S, \beta_0) \geq 1/2] \leq 2\delta$ by Step 1 at $\gamma = 1/2$.

Step 3 (Finitely many permutations). Fix the sample S and set $\zeta_b = \varphi^{(\pi_b)}(\beta_0)$. As each π_b is uniform and independent of S , the probability that replicate b rejects is exactly $V(S, \beta_0)$, so $\zeta_1, \dots, \zeta_B$ are i.i.d. Bernoulli with mean $V(S, \beta_0)$. Whenever $V(S, \beta_0) \leq \gamma$, the aggregated test can reject only if $\sum_b \zeta_b$ exceeds its mean by at least $B(1/2 - \gamma)$, so Hoeffding's inequality gives

$$\mathbb{P}_{H_0}[\tilde{\varphi}_B(\beta_0) = 1 \mid S] \leq \mathbb{P}\left[\sum_b \zeta_b \geq B/2 \mid S\right] \leq e^{-2B(1/2-\gamma)^2}.$$

Averaging over S gives the second bound. The display applies to every sample with $V(S, \beta_0) \leq \gamma$, and the remaining samples occur with probability at most δ/γ by Step 1. $\square$

For odd B the majority vote coincides with comparing a single aggregated statistic to the unchanged threshold.

Proposition 7.18 (Median-statistic form). *For odd B ,*

$$\tilde{\varphi}_B(\beta_0) = 1 \iff \text{med}\left(|\widetilde{W}^{(\pi_1)}(\beta_0)|, \dots, |\widetilde{W}^{(\pi_B)}(\beta_0)|\right) > \tau_n(\delta).$$

Proof. For odd B the median of B numbers is their $\frac{B+1}{2}$ th smallest. It exceeds the rejection threshold $\tau_n(\delta)$ exactly when fewer than $\frac{B+1}{2}$ of the numbers are at most $\tau_n(\delta)$. That is, when at least $\frac{B+1}{2} > B/2$ of them exceed $\tau_n(\delta)$. Since $|\widetilde{W}^{(\pi_b)}(\beta_0)| > \tau_n(\delta)$ is precisely the rejection $\varphi^{(\pi_b)}(\beta_0) = 1$, this is the event $\sum_b \varphi^{(\pi_b)}(\beta_0) > B/2$. $\square$

Inverting the aggregated test produces a confidence set, and the geometry of the geometry of Section 7.4 survives the aggregation.

Corollary 7.19 (Aggregated confidence set). *Let $CS_{1-\delta}^{(b)}$ denote the confidence set (33) computed on $S^{(\pi_b)}$ and define*

$$\widetilde{CS}_B := \{\beta_0 \in \mathbb{R} : \widetilde{\varphi}_B(\beta_0) = 0\} = \left\{ \beta_0 \in \mathbb{R} : \beta_0 \in CS_{1-\delta}^{(b)} \text{ for at least half of } b = 1, \dots, B \right\}.$$

Then, for every $\gamma \in (0, 1/2)$,

$$\mathbb{P}_\beta[\beta \in \widetilde{CS}_B] \geq 1 - \frac{\delta}{\gamma} - e^{-2B(1/2-\gamma)^2}, \quad (38)$$

and $\widetilde{CS}_B$ is a finite union of at most $kB + 1$ closed intervals, whose endpoints lie among the at most $2kB$ candidate points obtained by applying Corollary 7.5 to each of the B reordered samples.

write proof

8 Simulation

rewrite and clean up

8.1 Data Generation

Data is simulated from

$$Y = \beta X + \varepsilon_Y$$

where X is generated by

$$X = \mu_{ZX}Z + \varepsilon_X$$

with $Z \sim N(0, 1)$. In order to test the models under different regimes, we utilise three different shock families for ε_Y : Normal ($\sim N(0, 1)$), t ($\sim t(\nu)$), and centred Pareto ($\sim \text{Pareto}(\alpha)$). The base values are: $\beta = 1, \mu_{ZX} = 1, \sigma_{ZX}^2 = 2.5, \sigma_{Z\varepsilon}^2 = 1, \rho = 0.5, \delta = 0.05$, and $n = 2000$ unless stated otherwise. At this n , all instrument strength conditions hold.

8.2 Competitor Estimators

Besides the standard Mean IV estimator of Section 3, we include Catoni's M-estimator of the mean (Catoni, 2012) as a competitor. Alongside Median-of-Means, it is the other canonical sub-Gaussian mean estimator under the sole assumption of finite variance (Devroye et al., 2016), which makes it the natural robust benchmark for the estimators of Sections 4–5.

For a sample $(V_i)_{i=1}^n$ with mean μ_V and variance σ_V^2 , Catoni's estimator $\hat{\mu}_V^{\text{Cat}}$ is the root of

$$\sum_{i=1}^n \psi(\alpha(V_i - \theta)) = 0, \quad \psi(u) = \text{sign}(u) \ln\left(1 + |u| + \frac{u^2}{2}\right),$$

where the bounded influence function ψ replaces the identity implicit in the sample mean, so that no single observation can move the estimate by more than $O(1/\alpha)$. The tuning parameter α is chosen from the target confidence level and the noise scale as

$$\alpha = \sqrt{\frac{2 \ln(2/\delta)}{nv \left(1 + \frac{2 \ln(2/\delta)}{n-2 \ln(2/\delta)}\right)}},$$

which, for any variance bound $v \geq \sigma_V^2$, yields a sub-Gaussian deviation guarantee at confidence level $1 - \delta$ (Catoni, 2012). In the simulations v is not treated as known: it is replaced by a preliminary Median-of-Means estimate of the variance, so that the tuning itself remains valid under heavy tails and the comparison does not favour Catoni through oracle knowledge of scale.

The Mean IV analogue mirrors the Ratio-of-Medians construction: both cross-moments are estimated by Catoni's method and the ratio is taken,

$$\hat{\beta}_{\text{Cat}} = \frac{\hat{\mu}_{ZY}^{\text{Cat}}}{\hat{\mu}_{ZX}^{\text{Cat}}},$$

with the confidence budget split evenly across the two coordinates, i.e. each estimate is tuned at level $\delta/2$, matching the allocation used by the Ratio-of-Medians estimator in Theorem 4.1. Since the root equation is strictly decreasing in θ , each estimate is computed exactly by bisection.

We include a second competitor, the trimmed mean of Oliveira and Orenstein (2019) and Lugosi and Mendelson (2021), which reaches the same sub-Gaussian guarantee by truncation rather than by a bounded influence function. The sample is split into two halves $\mathcal{S}_1$ and $\mathcal{S}_2$. The first supplies the empirical quantiles at levels η and $1 - \eta$, with trimming level $\eta = 8 \ln(4/\delta)/(3n)$, and the second is truncated to the resulting window $[a, b]$ before being averaged,

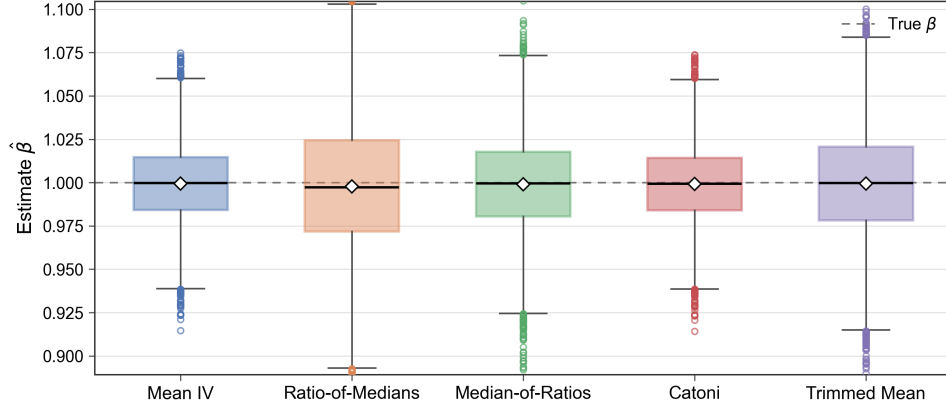
$$\hat{\mu}_V^{\text{Trim}} = \frac{2}{n} \sum_{i \in \mathcal{S}_2} \phi_{a,b}(V_i), \quad \phi_{a,b}(u) = \min\{\max\{u, a\}, b\}.$$

Splitting the sample is what makes the argument work, since the truncation window is then independent of the points it is applied to, and yields $|\hat{\mu}_V^{\text{Trim}} - \mu_V| \leq C\sigma_V \sqrt{\ln(4/\delta)/n}$ with probability $1 - \delta$ under nothing beyond a finite variance (Lugosi & Mendelson, 2021).

No scale enters the tuning. The truncation window consists of empirical quantiles of the data, so unlike Catoni's estimator the trimmed mean is scale-equivariant by construction and needs no preliminary variance estimate to be feasible. The IV analogue is formed exactly as for Catoni,

$$\hat{\beta}_{\text{Trim}} = \frac{\hat{\mu}_{ZY}^{\text{Trim}}}{\hat{\mu}_{ZX}^{\text{Trim}}},$$

with the confidence budget split evenly across the two coordinates and one split of the sample shared by both. Averaging over half of the observations costs a factor $\sqrt{2}$ in efficiency relative to the estimators that use all n .



Note. 10000 replications; $n = 2000$, Gaussian errors, $\mu_{ZY} = 1$, $\rho = 0.5$, $\delta = 0.05$. Axis truncated at 0.5/99.5% quantiles (0.15% of points outside).

Figure 1: Boxplot showing the cost of robustness under Gaussian errors

8.3 Simulation Results

8.3.1 Cost of Robustness under Gaussian Errors

As can be seen in Figure 1, all four estimators are centred on β , but their spreads differ. Mean IV and Catoni both have a standard deviation of 0.0224, MoR of 0.0274, and RoM 0.0396. These results are exactly as expected; under light tails, the sample mean is already the efficient estimator, so blocking and taking medians only throws information away. Catoni performs identically to Mean IV because its influence function is nearly linear on well-behaved data. That is, the Catoni estimator is the mean until extreme observations appear.

8.3.2 Benefit of Robustness under Heavy Tails

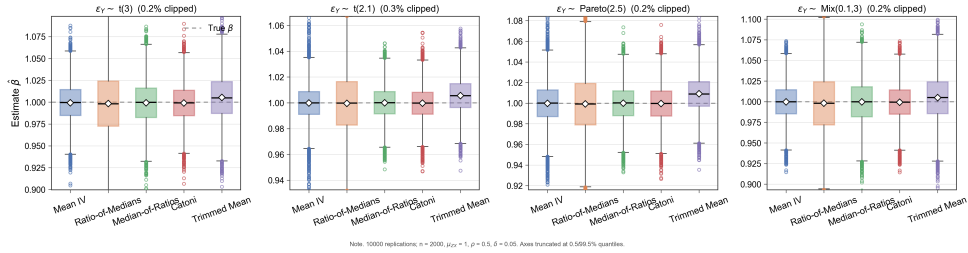


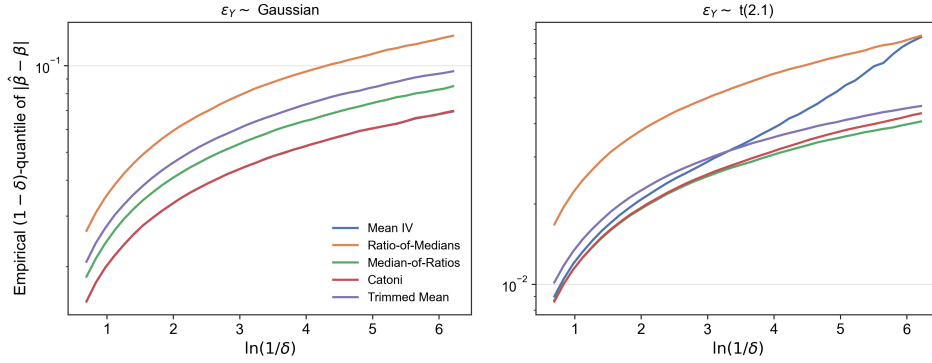
Figure 2: Boxplot showing the benefit of robustness under $t(3)$, $t(2.1)$, and $Pareto(2.5)$ tails

As per Figure 2, the benefit of MoM-type estimators becomes clear under $t(2.1)$ errors; the RMSE is 0.0128 for MoR and 0.0130 for Catoni against 0.0189 for Mean IV. The $(1 - \delta)$ error quantile is 0.0341 under MoR and 0.0446 under Mean IV. RoM is clearly insufficient in all cases.

The IQR of the Mean IV and MoR estimators is very similar; the difference between the two is in the outliers, with MoR being much less susceptible to extreme values. This is expected, as MoM-type estimators do not promise variance bounds, but tail bounds, so the benefit lives in the tails. The Pareto panel shows no bias for any of the estimators, showing that all estimators can deal with asymmetric distributions.

Catoni and MoR show very similar results but, visually, MoR seems to slightly improve on Catoni's tails. This conclusion is only drawn from a visual inspection of the graph. Even if both estimator yielded the same simulation results, the parsimony of the MoR estimator may make it more appealing if the data is known to be heavy tailed.

8.3.3 δ -dependence

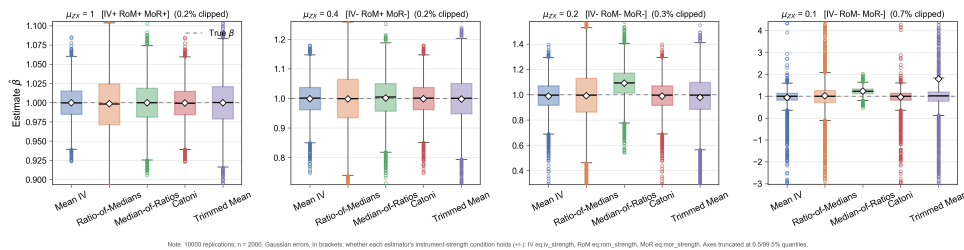


Note. 50000 replications; $n = 2000$. Log scale. A sub-Gaussian estimator grows like $\sqrt{\ln(1/\delta)}$ (flat); the polynomial rate of the empirical mean grows like $\delta^{-1/2}$ (steep at large $\ln(1/\delta)$).

Figure 3: Graph showing the empirical $(1 - \delta)$ -quantile of $|\hat{\beta} - \beta|$ against $\ln(1/\delta)$. Catoni and Mean IV have a near identical graph in the Normal regime

Figure 3 shows that, for the Normal panel, all four curves are parallel and gently concave, meaning that every estimator is effectively sub-Gaussian, with Mean IV and Catoni sitting the lowest at every δ . In the $t(2.1)$ regime we can see that the Mean IV curve becomes convex at around $\ln(1/\delta) = 4$ which, as stated in Remark 3.4, is unavoidable. The more confidence is demanded, the more MoM wins when compared against Mean IV.

8.3.4 Instrument Strength Conditions



Note. 10000 replications; $n = 2000$. Gaussian errors. In brackets: whether each estimator's instrument strength condition holds (+/-); IV eq IV_strength, RoM eq RoM_strength, MoR eq MoR_strength. Axes truncated at 0.5/99.5% quantiles.

Figure 4: Boxplot showing the effect of weaker instrument strength conditions with a sweeping μ_{ZX}

Figure 4 shows that, at $\mu_{ZX} = 0.4$, where MoM conditions fail but the Mean IV condition holds, every estimator is still centred, but Mean IV and Catoni are the tightest. As the instrument strength condition weakens, Mean IV and Catoni stay centred on β , but produce unreliable estimates, with a standard deviation of 2.5 and minimum and maximum values in the hundreds. MoR, though, remains stable, albeit highly biased, with a mean estimate of 1.23. On RMSE, MoR gives the best fit here, with a value of 0.28, against 2.5 of Mean IV and 2.8

of Catoni. The RoM estimator performs worse in all cases. Under weak instruments, IV, RoM, and Catoni fail by variance explosion, and MoR fails by bias.

8.3.5 Against extreme distributions

Decide whether to include

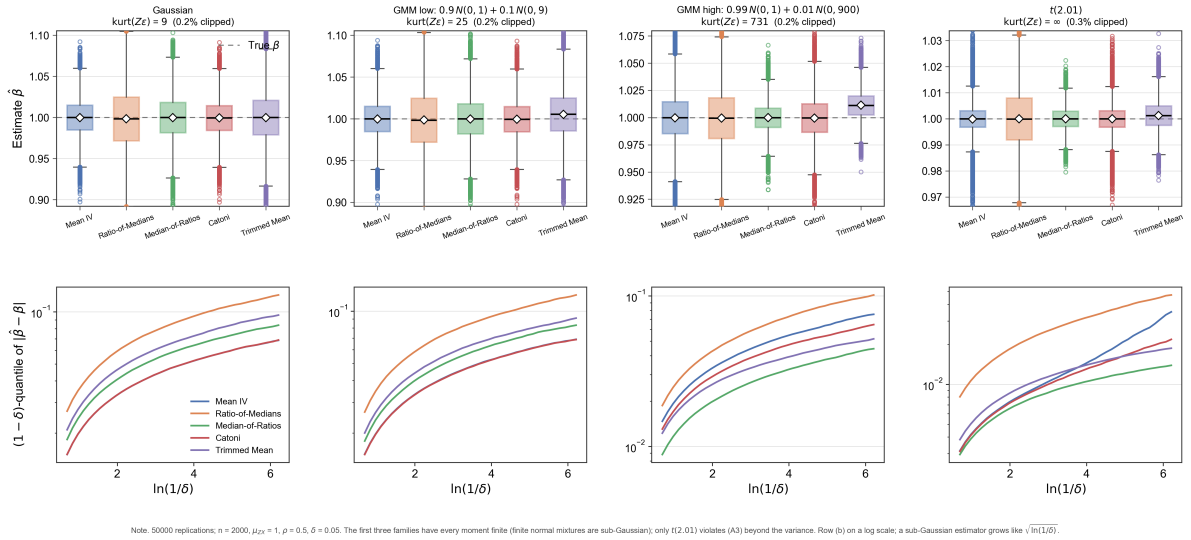


Figure 5: Sampling distributions (top) and empirical $(1 - \delta)$ -quantiles of $|\hat{\beta} - \beta|$ (bottom) for three families with every moment finite against $t(2.01)$. The mixtures are Gaussian, so their tails are sub-Gaussian however large the kurtosis

Figure 5 separates heavy tails from mere by contrasting two normal mixtures, which have every moment finite, against $t(2.01)$. Since the data generated from a GMM will have all moments finite, the estimators used will all be sub-gaussian, this exercise is only conducted to test it against a Tukey contaminated model, inspired by Tukey (1960). MoR performs best in the most extreme case, where a small proportion of very large outliers exist. The final panel compares against an even more extreme $t(2.01)$. MoR beats Catoni convincingly, but the benefit over the Trimmed Mean estimator is marginal

8.3.6 Empirical Size

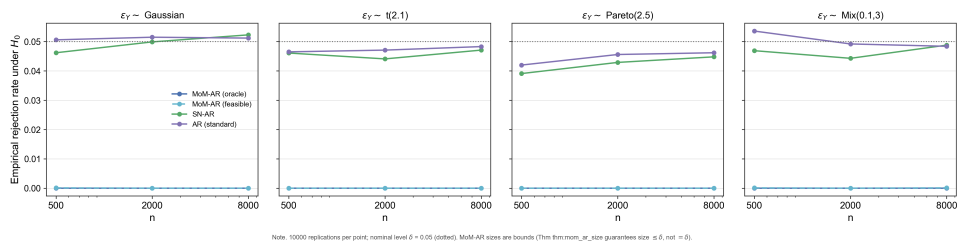


Figure 6: Plot of the size of all tests

Figure 6 shows that, at the true β , both the oracle and feasible MoM-AR test reject in at most 0.01% of 10,000 replications in every design, against the nominal 5%. The finite-sample

size bound of Theorem 7.1 holds and is very conservative; the oracle threshold τ_n is 7.9 times the actual standard deviation of the statistic under H_0 in the Gaussian design at $n = 2000$, and 16.7 times under $t(2.1)$; it is built from Chebyshev and Hoeffding distribution-agnostic worst-case scenarios, so it is fixed by $\sigma_{Z\varepsilon}$ alone and cannot adapt to the tails. The feasible threshold, rebuilt from a robust scale estimate, instead stays between 7.0 and 7.3 times that standard deviation in every design and at every sample size.

Comparing the AR tests with each other, both are close enough to the nominal 5% level value, though as the distributions get heavier tails ($t(2.1)$ and *Pareto*(2.5)), they become slightly more conservative. It is worth noting that the self-normalised AR test is consistently slightly more conservative than the standard AR test.

8.3.7 Power of the Four Tests

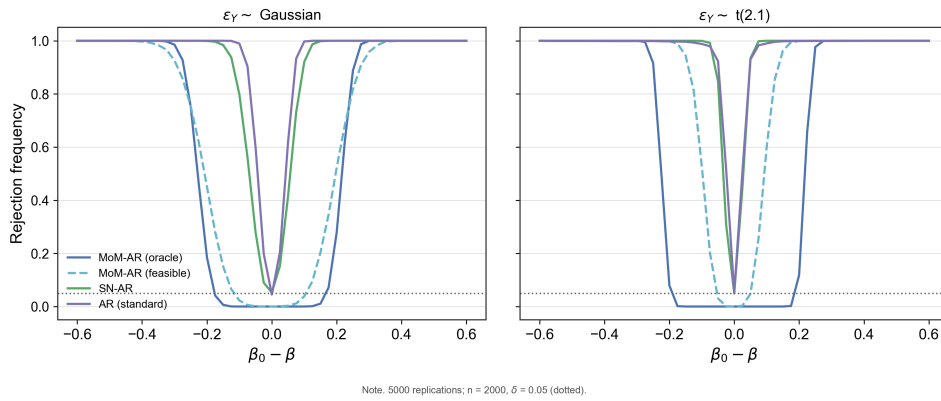


Figure 7: Power curves of all tests

Figure 7 shows that the MoM-AR tests are less powerful than the AR-based tests, with the feasible version being slightly more powerful than the oracle version in the Normal setting. In the heavy tail setting, the feasible MoM-AR test is much more powerful. This is due to the MAD-based scale estimate setting $\hat{\sigma}_{Z\varepsilon}^{\text{MAD}}$ below the true $\sigma_{Z\varepsilon}$. This is not expected; it is a cancellation specific to the DGP, not a guarantee. The standard AR test is more powerful than the self-normalised AR test in the Normal setting, though this benefit decreases as tails become larger, as can be seen in the $t(2.1)$ case, where the two are nearly indistinguishable. If the finite-sample guarantee is needed, one has to pay in power, though, as can be seen by comparing the self-normalised to the standard AR, the cost is not that high.

8.3.8 Partition Randomness

Every procedure in Sections 4–7 partitions the sample into k blocks. The proofs require only that the partition be fixed in advance of the data, so which rows are grouped together is irrelevant to the guarantee. It is not irrelevant to the number reported, since samples usually arrive ordered by date, region, or cohort, and blocks cut from the file as it stands would then differ for reasons unrelated to sampling noise. We therefore draw a uniformly random permutation of the rows and cut consecutive blocks from the permuted sample, which makes the estimate independent of the file order at the cost of making it a function of the seed.

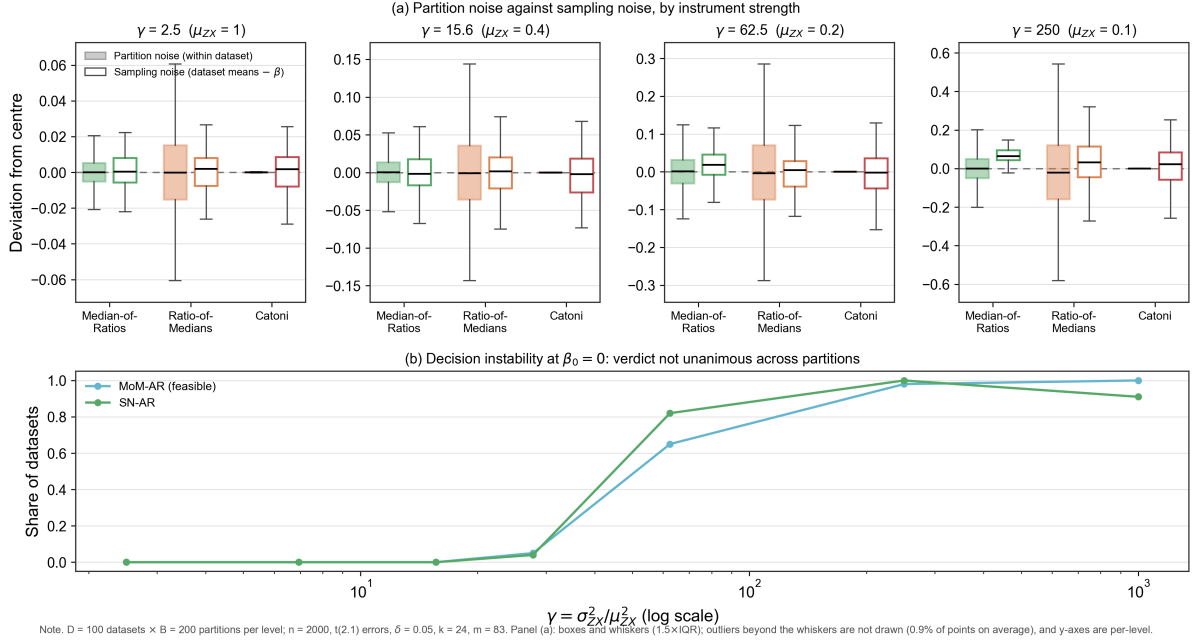


Figure 8: Partition noise against sampling noise, and the share of datasets whose test verdict at $\beta_0 = 0$ depends on the partition, against instrument strength $\gamma = \sigma_{ZX}^2 / \mu_{ZX}^2$

The partition introduces noise. At $\gamma = 2.5$, where all strength conditions hold, the partition accounts for 34% of the total variance of the Median-of-Ratios estimator and for 78% of that of Ratio-of-Medians. Catoni's estimator is effectively partition-free, as the blocking enters it only through the variance pre-estimate. As the instrument weakens both block estimators deteriorate, and the deterioration is mostly in the partition; the Median-of-Ratios share rises to 93% at $\gamma = 1000$, so that at that point the reported number is more a function of the seed than of the sample.

Panel (b) asks what this costs a conclusion rather than an estimate. It reports the share of datasets on which rejecting the null hypothesis depends on the seed. Up to $\gamma \approx 16$ the verdict is unanimous on every dataset, so the partition has no effect on inference. Instability then rises steeply, to 65% of datasets for the feasible MoM-AR test and 82% for the self-normalised test at $\gamma = 62.5$, and to essentially every dataset by $\gamma = 250$.

Figure 9 shows the same phenomenon from the position of a researcher holding one data set. Each panel fixes a single sample and redraws the partition 2,000 times, so all of the spread displayed is due to the seed alone, with the partition-free Mean IV estimate marked for reference. Under a strong instrument the two block estimators concentrate near it, with Ratio-of-Medians already visibly the wider of the two. At $\gamma = 1000$ Ratio-of-Medians ranges over roughly $[-5, 10]$ on a single dataset, while Median-of-Ratios remains a narrow spike, displaced from β but stable.

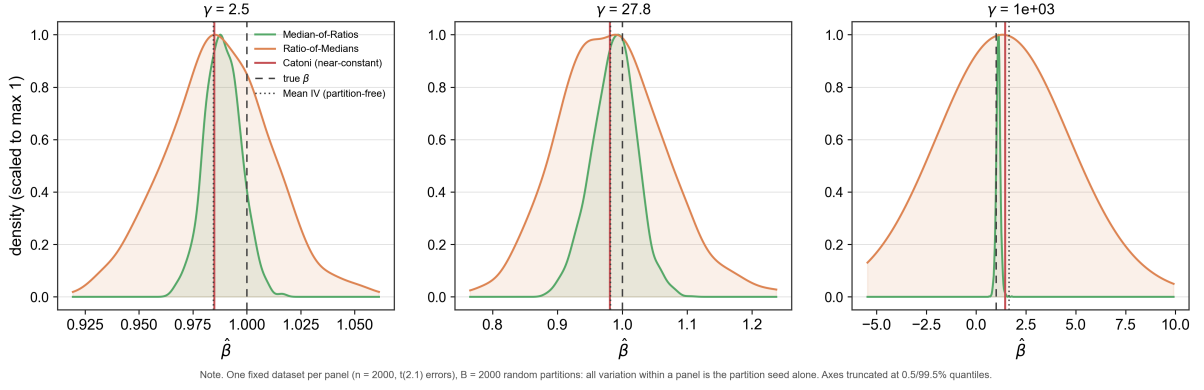


Figure 9: Estimator distributions over random partitions of a single fixed dataset, at three instrument strength levels

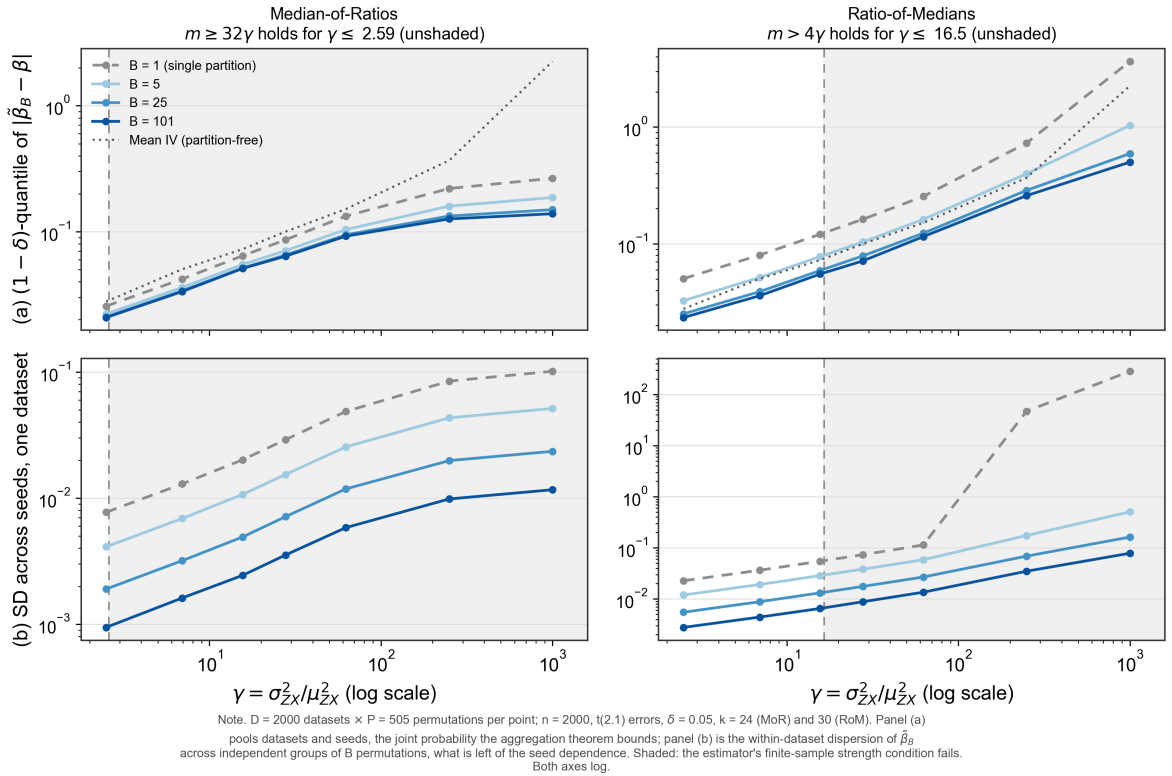


Figure 10: Aggregated Median-of-Ratios and Ratio-of-Medians estimators against instrument strength γ , by aggregation depth B : (a) the $(1 - \delta)$ -quantile of $|\hat{\beta}_B - \beta|$, (b) the dispersion of $\hat{\beta}_B$ across seeds on a fixed data set

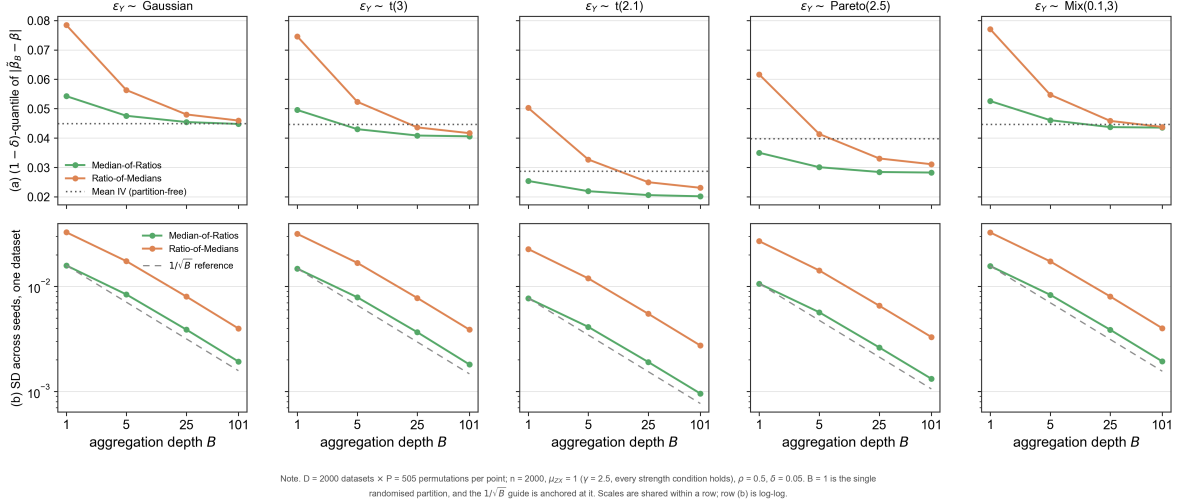


Figure 11: The same two quantities against aggregation depth B under five error families, at the canonical strong-instrument design

8.3.9 Aggregated Estimation

Expand: the seed dispersion of panel (b) falls by a factor 8.1–8.7 from $B = 1$ to $B = 101$ at every γ and under every error family, against the $\sqrt{101} = 10.05$ a mean would give; the gap is the efficiency factor $\sqrt{\pi/2}$ of the median. Accuracy never pays for it: the deviation quantile of panel (a) falls monotonically in B , by 1.2–1.9 times for Median-of-Ratios and 2.2–7.3 times for Ratio-of-Medians, and the failure rate at the single-partition tolerance drops from δ to 0.014 or below, so the factor 2 in δ of Theorem 6.5 is an artefact of the Markov step and not a cost. The gain is largest exactly where the strength conditions fail — Ratio-of-Medians at $\gamma = 1000$ goes from a seed SD of 284, a median block denominator near zero, to 0.08 — and both panels have essentially saturated by $B = 25$. The error family barely matters in Figure 11, the aggregation acting on the partition rather than on the tails, and the ranking of the two estimators is the one of Section 8.3.2 unchanged.

8.3.10 Aggregated Inference

Expand: size is identically 0 at every γ and B , so Figure 12 may be better as one sentence; power gains up to 0.15 on the shoulders, saturating by $B = 25$.

9 Case Studies

The simulations of Section 8.3 evaluated the estimators and tests on data generated to satisfy (A1)–(A3) by construction, with the instrument strength conditions holding or failing by design. This section applies the same procedures to two instrumental variable studies, where neither the tail behaviour nor the instrument strength is under our control.

The two applications are chosen to complement each other regarding instrument strength. Angrist and Krueger (1991) instrument years of schooling with quarter of birth. There μ_{ZX} is

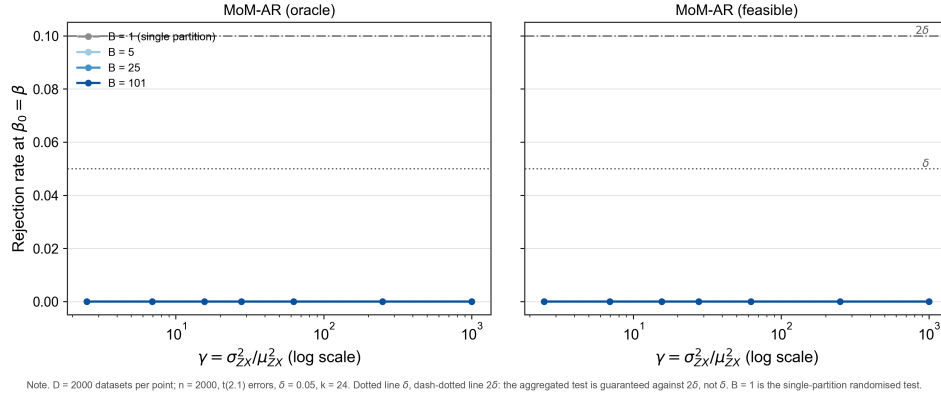


Figure 12: Empirical size of the aggregated MoM-AR test at $\beta_0 = \beta$ against instrument strength γ , by aggregation depth B

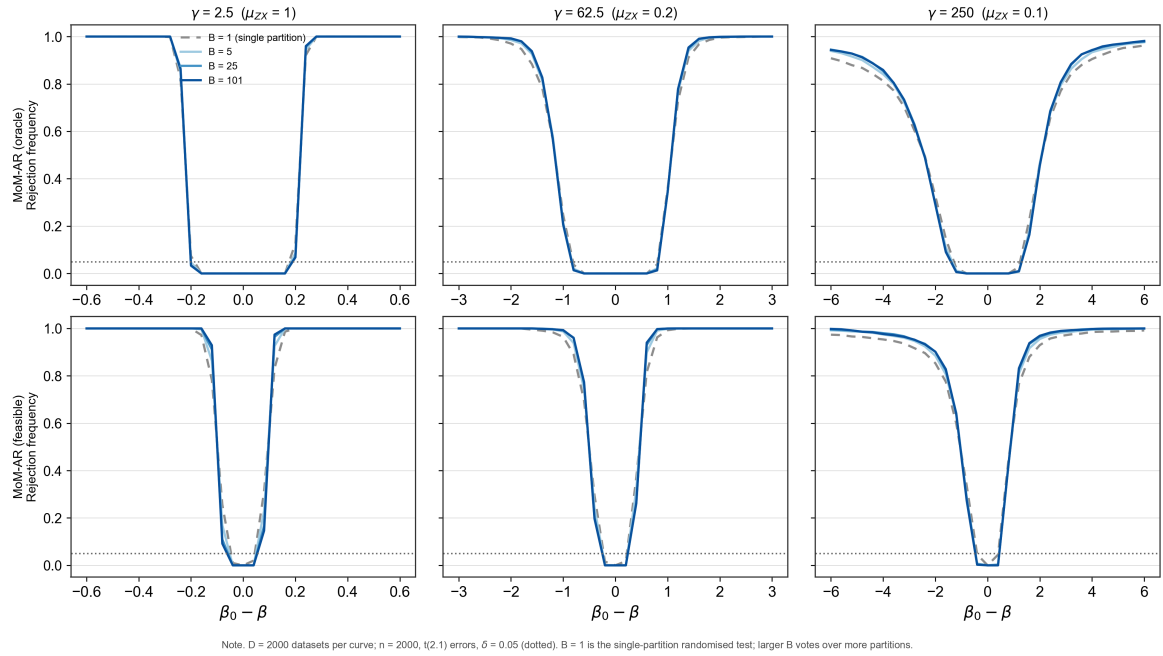


Figure 13: Power curves of the aggregated MoM-AR test at three instrument strength levels, by aggregation depth B

small relative to σ_{ZX} and every finite-sample strength condition of Sections 3–5 fails. Angrist and Evans (1998) instrument family size with the sex composition of the first two children. Their instrument is stronger relative to the noise and all three strength conditions hold. Reading the two studies together therefore separates what the estimators do when the theory applies from what they do when it does not.

The purpose of this exercise is not to check whether the MoM-based estimators recover the published numbers, but rather to check what these methods buy, and cost, on real data.

9.1 Empirical Strategy

9.1.1 Mapping the applications into the scalar model

The theory developed concerns $Y_i = \beta X_i + \varepsilon_i$ with a single instrument Z_i and $\beta = \mu_{ZY}/\mu_{ZX}$. Both applications are therefore taken in their Wald form, that is, the just-identified specification with a single binary instrument and no covariates. In each case the published paper reports this specification alongside its covariate-adjusted counterparts, which makes the replication target unambiguous: Panel B of Table III in Angrist and Krueger (1991) and columns (1)–(2) of Table 5 in Angrist and Evans (1998).

Both papers write their models with an intercept. Our methods have been developed without an intercept, so we demean the three variables at the full-sample level before applying any procedure,

$$Y_i \leftarrow Y_i - \bar{Y}, \quad X_i \leftarrow X_i - \bar{X}, \quad Z_i \leftarrow Z_i - \bar{Z}.$$

With a binary instrument the sample ratio $\hat{\mu}_{ZY}/\hat{\mu}_{ZX}$ of the demeaned variables is then algebraically identical to the Wald estimator $(\bar{Y}_1 - \bar{Y}_0)/(\bar{X}_1 - \bar{X}_0)$, so the Mean IV estimator of Section 3 reproduces the published point estimate. Demeaning also keeps the moment $W_i(\beta_0) = Z_i(Y_i - \beta_0 X_i)$ of Section 7.1 centred at zero under H_0 , which would not hold if the intercept were left in.

The interpretation of the MoM-based estimators are the same as the Wald estimators; the estimands capture the local average treatment effect since they estimate the same ratio of population moments.

9.1.2 Blocking, and why the partition is randomised

9.1.3 Reported quantities

For each application we report, at $\delta = 0.05$ throughout: the replication of the published table; the four point estimators (Mean IV, Ratio-of-Medians, Median-of-Ratios, and the Catoni competitor of Section 8.2); the three feasible confidence sets (MoM-AR with the robust scale estimate, the self-normalised AR test, and the standard AR test as a baseline); and a plug-in check of the finite-sample strength conditions. The oracle MoM-AR test of Theorem 7.1 is not feasible on real data, since $\sigma_{Z\varepsilon}$ is unknown, so only its feasible counterpart as developed earlier appears here.

9.2 The Causal Effect of Compulsory School Attendance on Earnings

9.2.1 Data and replication

Angrist and Krueger (1991) exploit the fact that compulsory schooling laws are measured with regards to age, and not grade completed. Therefore, children born earlier in the calendar year reach the legal dropout age having completed less schooling, so quarter of birth shifts educational attainment without, the authors argue, affecting earnings through any other channel. We replicate Panel B of their Table III, the 1980 Census sample of men born 1930–1939, with Y the log weekly wage, X years of education and Z an indicator for birth in the first quarter.

Table 1 reproduces the published group means, differences and Wald estimate. The posted extract contains 329,509 observations against the 327,509 reported in the paper, as it is the sample used for their Table V; every reported quantity nonetheless matches to the precision printed in the original.

Why is there a 2000 observation discrepancy?

Table 1: Replication of Angrist and Krueger (1991), Table III, Panel B (1980 Census, men born 1930–1939). The posted extract has $n = 329,509$ observations against the published 327,509; standard errors in parentheses.

	Replicated		Published	
	Estimate	(s.e.)	Estimate	(s.e.)
ln(wkly. wage), born Q1	5.8916		5.8916	
ln(wkly. wage), born Q2–4	5.9027		5.9027	
difference	-0.01110	(0.00274)	-0.01110	(0.00274)
Education, born Q1	12.6881		12.6881	
Education, born Q2–4	12.7969		12.7969	
difference	-0.1088	(0.0133)	-0.1088	(0.0132)
Wald return to education	0.1020	(0.0240)	0.1020	(0.0239)
OLS return to education	0.0709	(0.00034)	0.0709	(0.0003)

Table 2: Robust IV estimates and 95% confidence sets for the return to education, AK91 extract ($\delta = 0.05$; demeaned Y , X , Z ; $k = 24$, $m = 13,729$). The MoM-AR test uses the feasible robust scale estimate $\hat{\sigma}_{Z\varepsilon} = 0.3429$; the oracle version is infeasible on real data.

Estimator	$\hat{\beta}$
Wald / Mean IV	0.1020 (0.0240)
Ratio-of-Medians ($k = 30$)	0.1023
Median-of-Ratios ($k = 24$)	0.0584
Catoni ratio ($\delta/2$ per coordinate)	0.1018
Test	95% confidence set
MoM-AR (feasible)	$[-0.1923, 0.3839]$
SN-AR	$[-0.0097, 0.1744] \cup [0.1761, 0.1935] \cup [0.2221, 0.2325]$
AR (standard)	$[0.0554, 0.1525]$

9.2.2 Instrument strength

The plug-in moments of the demeaned data are $\hat{\mu}_{ZX} = -0.0203$ and $\hat{\sigma}_{ZX}^2 = 2.031$, so $\hat{\sigma}_{ZX}^2/\hat{\mu}_{ZX}^2 \approx 4,900$. Every finite-sample condition of the point estimators fails. The standard IV condition (7) requires $n \geq 789,667$ against the 329,509 available; the Median-of-Ratios condition (17) requires a block size $m \geq 157,933$ against $m = 13,729$; and the Ratio-of-Medians condition (14) requires $m > 19,742$ against $m = 10,983$.

9.2.3 Robust estimates and confidence sets

mention seed chosen is “1971”

Table 2 reports the four estimators and the three confidence sets for one documented partition.

The Catoni and Ratio-of-Medians estimators both agree closely with the Wald estimator, while the Median-of-Ratios sits below them at this seed. The feasible MoM-AR set is far wider than the standard AR set and includes. The self-normalised AR test is a union of three intervals, one of which also includes zero. Additionally, none of the confidence sets reject the Wald value.

The consequence is clear; the point estimate of Angrist and Krueger (1991) is reproduced exactly, and nothing here contradicts their conclusion. The finite-sample robust tests are simply uninformative in this application. Their confidence sets are wide enough to contain both zero and the Wald estimate, which is the price.

9.2.4 Partition sensitivity

Table 3 and Figure 14 summarise $B = 1,000$ independent partitions of the same data.

The dispersion induced by the partition is of the same order as the sampling uncertainty of the Wald estimator: the Median-of-Ratios estimate has a standard deviation of 0.019 across partitions and ranges over $[0.028, 0.150]$, against a Wald standard error of 0.024. As stated earlier, Catoni’s estimator is essentially partition free. The block means $\hat{\mu}_{ZX}^{(j)}$ share a sign in only 30.4% of partitions, so the sufficient condition of Proposition 7.7 usually fails, and the self-normalised confidence set is a union of several intervals in 14.6% of draws. The feasible MoM-AR set excludes zero in 1.8% of partitions against 91.6% for the self-normalised set and 100% for the partition-free standard AR test.

Discuss consequence of how sensitive inference is on seed

9.2.5 Aggregated estimates and confidence sets

Table 4 applies the aggregation of Section 6 to the $B = 1,000$ partitions of the previous subsection, reporting the aggregated estimator of Definition 6.1 and the aggregated confidence set of Corollary 7.19. Every permutation drawn enters the same aggregate, so the table has no seed left to report, and it is directly comparable with Table 2, which is the same quantities on one partition.

Table 3: Partition sensitivity on the AK91 extract: $B = 1,000$ independent random partitions, each shared by all MoM-based procedures ($\delta = 0.05$). The partition-free Wald estimate is 0.1020. Top panel: dispersion of the point estimates across partitions. Bottom panel: rejection and shape frequencies of the 95% confidence sets; the standard AR test does not depend on the partition and is shown for reference.

Estimator	Median	SD	$Q_{0.05}$	$Q_{0.95}$	Range
Median-of-Ratios	0.0992	0.0189	0.0666	0.1284	[0.0280, 0.1496]
Ratio-of-Medians	0.1013	0.0211	0.0686	0.1373	[0.0409, 0.2036]
Catoni ratio	0.1018	0.0000	0.1018	0.1019	[0.1018, 0.1019]
Test	% rej. 0	% rej. β_{OLS}	% split	% unb.	Med. len.
MoM-AR (feasible)	1.8	0.0	0.1	0.0	0.431
SN-AR	91.6	10.1	14.6	0.0	0.145
AR (standard)	100.0	0.0	0.0	0.0	0.097

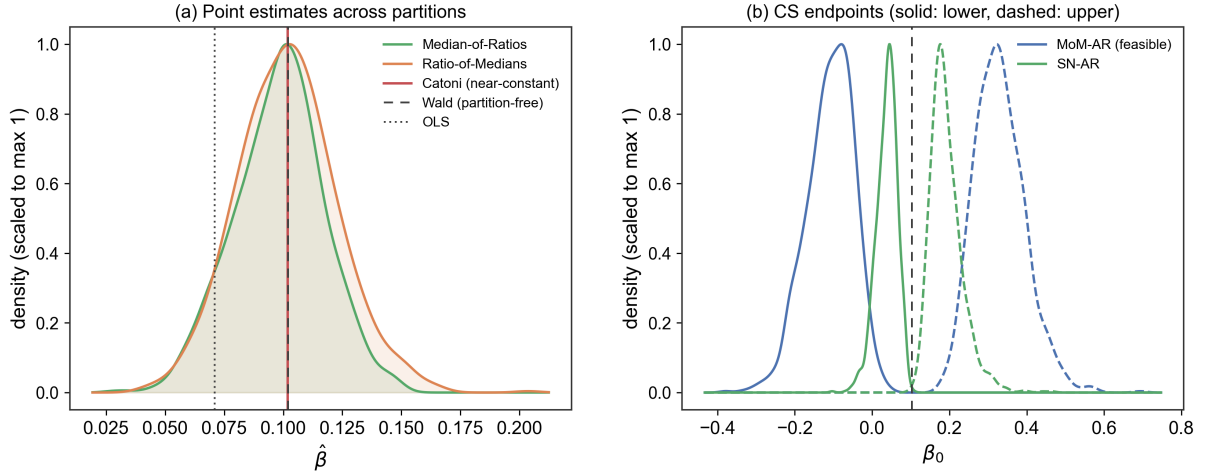


Figure 14: Partition sensitivity on the Angrist and Krueger (1991) extract over $B = 1,000$ random partitions. Panel (a): kernel density estimates of the point estimates, each scaled to unit maximum, with the partition-free Wald and OLS values marked. Panel (b): densities of the outer endpoints of the 95% confidence sets (solid: lower, dashed: upper).

Expand: Aggregation buys back most of what Section 9.2.4 showed the seed was costing. The Median-of-Ratios estimate moves from 0.0584 on the documented partition, one draw from a range of $[0.028, 0.150]$, to 0.0992, within 0.003 of the partition-free Wald value 0.1020; Ratio-of-Medians lands at 0.1013. The self-normalised set is the sharper illustration: on one partition it was a union of three intervals, and whether it excluded zero was a function of the seed in 8.4% of draws. Aggregated it is the single interval $[0.0413, 0.1829]$, it excludes zero, and there is no seed left to vary. Nothing is paid in width for that: the aggregated sets have essentially the median length of the per-partition sets, 0.142 against a median of 0.15 for the self-normalised test and 0.428 against 0.43 for MoM-AR. What aggregation does not repair is the conservativeness of $\tau_n(\delta)$, so the feasible MoM-AR set still contains both zero and the Wald estimate, exactly as in Table 2. Note also that the aggregated set is guaranteed at 2δ rather than δ , so a like-for-like comparison would run the underlying test at $\delta/2$ (Remark 6.6).

Table 4: Aggregated estimates and the aggregated 95% confidence set on the AK91 extract, over all $B = 1,000$ random partitions of Table 3 (Definition 6.1 and Corollary 7.19, $\delta = 0.05$; $k = 24$ for MoR and the tests, $k = 30$ for RoM). Table 2 is the same quantities on a single partition. The Wald estimator and the standard AR test do not depend on the partition and are shown for reference; by Corollary 7.19 the aggregated set is guaranteed at 2δ , not δ .

Aggregated estimator	$\tilde{\beta}_B$
Median-of-Ratios	0.0992
Ratio-of-Medians	0.1013
Wald / Mean IV (partition-free)	0.1020
Aggregated test	95% confidence set
MoM-AR (feasible)	$[-0.1031, 0.3245]$
SN-AR	$[0.0413, 0.1829]$
AR (standard, partition-free)	$[0.0554, 0.1525]$

9.3 The Causal Effect of Family Size on Parental Labour Supply

9.3.1 Data and replication

Angrist and Evans (1998) estimate the effect of childbearing on parents' labour supply, instrumenting an indicator for having more than two children with an indicator for the first two children having the same sex. Parents with two same-sex children are more likely to have a third, while the sex mix itself is as good as randomly assigned, which is the identifying claim. We replicate columns (1)–(2) of their Table 5, which is the 1980 Census sample of all women aged 21–35 with two or more children, for the five labour-supply outcomes.

Table 5 reports the replication. The reconstructed sample contains 394,840 observations against the 394,835 of the published table, a five-observation discrepancy of the posted archive also reported by Farbmacher, Guber and Vikström (2018), whose selection rules we follow. The first stage is 0.0595 against a published 0.0600, and all five Wald estimates match at printed precision.

Table 5: Replication of Angrist and Evans (1998), Table 5, columns (1)–(2): 1980 PUMS, all women aged 21–35 with two or more children, *Same sex* instrument, *More than 2 children* as endogenous regressor. Reconstructed sample $n = 394,840$ against the published 394,835 (five-observation archive discrepancy documented by Farbmacher et al. (2018)). Standard errors in parentheses; Wald SEs by the delta method.

	Replicated		Published	
	Estimate	(s.e.)	Estimate	(s.e.)
<i>More than 2 children</i>	0.0595	(0.0016)	0.0600	(0.0016)
<i>Number of children</i>	0.0754	(0.0026)	0.0765	(0.0026)
Worked for pay, difference	-0.0079	(0.0016)	-0.008	(0.0016)
Wald	-0.132	(0.026)	-0.133	(0.026)
Weeks worked, difference	-0.3787	(0.0709)	-0.3826	(0.0709)
Wald	-6.360	(1.181)	-6.38	(1.17)
Hours/week, difference	-0.3089	(0.0602)	-0.311	(0.0602)
Wald	-5.187	(1.006)	-5.18	(1.0)
Labor income, difference	-130.3	(34.4)	-132.5	(34.4)
Wald	-2188.3	(573.6)	-2208.8	(569.2)
ln(Family income), difference	-0.0016	(0.0041)	-0.0018	(0.0041)
Wald	-0.026	(0.069)	-0.029	(0.068)

Table 6: Robust IV estimates and 95% confidence sets, AE98 1980 all-women sample ($\delta = 0.05$; demeaned Y , X , Z ; one shared random partition, $k = 24$, $m = 16,451$ for MoR and the tests, $k = 30$ for RoM). The MoM-AR test uses the feasible robust scale estimate; the oracle version is infeasible on real data.

Outcome	Wald/Mean IV	MoR	RoM	Catoni
Worked for pay	-0.1318	-0.1118	-0.1398	-0.1318
Weeks worked	-6.3596	-6.2532	-7.7001	-6.3595
Hours/week	-5.1872	-4.8504	-6.6122	-5.1871
Labor income	-2188.3	-1734.1	-2860.9	-2188.4
ln(Family income)	-0.0262	-0.0436	0.0481	-0.0262
Outcome	MoM-AR (feasible)	SN-AR	AR (standard)	
Worked for pay	[−0.307, 0.072]	[−0.170, −0.051]	[−0.184, −0.080]	
Weeks worked	[−18.46, 5.48]	[−10.32, −2.47]	[−8.68, −4.04]	
Hours/week	[−14.88, 3.77]	[−8.72, −2.49]	[−7.16, −3.22]	
Labor income	[−5013, 1281]	[−2764, −683]	[−3313, −1061]	
ln(Family income)	[−1.104, 1.194]	[−0.362, 0.262]	[−0.161, 0.109]	

9.3.2 Instrument strength

Here $\hat{\mu}_{ZX} = 0.0149$ and $\hat{\sigma}_{ZX}^2 = 0.0598$, so $\hat{\sigma}_{ZX}^2/\hat{\mu}_{ZX}^2 \approx 270$, roughly eighteen times smaller than in Section 9.2.2. The standard IV condition (7) requires $n \geq 43,219$ and is met with 394,840 observations; the Median-of-Ratios condition (17) requires $m \geq 8,644$ and is met with $m = 16,451$; the Ratio-of-Medians condition (14) requires $m > 1,080$ and is met with $m = 13,161$.

9.3.3 Robust estimates and confidence sets

Table 6 reports the four estimators and the three confidence sets for one documented partition. The estimators agree closely for the bounded outcomes and less closely for the unbounded ones. For Labour income the Median-of-Ratios estimate is 21% below the Wald estimate and the Ratio-of-Medians estimate 31% above it, and for $\ln(\text{Family income})$ the latter reverses sign. The ordering follows the tail behaviour of Section ??, but each gap lies within roughly two partition standard deviations of Table 7, so it reflects the draw of the partition rather than the estimator.

None of the five feasible MoM-AR sets excludes zero, and all are three to five times longer than the standard AR sets. This is not a symptom of a weak instrument, since all three strength conditions hold here; at $\delta = 0.05$ the threshold $\tau_n(\delta)$ of (31) exceeds the classical critical value by a factor $\sqrt{32\ln(1/\delta)}/1.96 \approx 5$, which accounts for essentially all of the gap. The binding constraint is the conservativeness of the threshold, not instrument strength. The self-normalised sets exclude zero for all four labour-supply outcomes and are each a single interval, in contrast with Section 9.2, where the same test returned a union of three; for $\ln(\text{Family income})$ no procedure excludes zero, matching the null result of Angrist and Evans (1998).

9.3.4 Partition sensitivity

Table 7 and Figures 15 and 16 summarise $B = 1,000$ partitions of the same data. For every outcome the estimator distributions are centered on the partition-free Wald value, and the partition standard deviation is between 68% and 71% of the sampling standard error, 0.018 against 0.026 for Worked for pay. No confidence set, for any outcome or any partition, is split or unbounded. The self-normalised test excludes zero in 99% or more of partitions for the three labour-supply outcomes and in 90.4% for Labour income, the difference being visible in panel (b), where the density of the upper endpoint sits clear of zero in the first case and straddles it in the second.

The contrast with Section 9.2.4 is not one of magnitude, since the dispersion there is a comparable fraction of the sampling standard error. In Angrist and Krueger (1991) the partition determines the shape of the confidence set and moves the Median-of-Ratios estimate over $[0.028, 0.150]$, so the reported answer depends on a seed. Here, every partition delivers the same verdict for every outcome.

9.3.5 Aggregated estimates and confidence sets

Table 8 applies the same aggregation to the $B = 1,000$ partitions, for each of the five outcomes, against the single-partition Table 6.

Table 7: Partition sensitivity on the AE98 1980 all-women sample: $B = 1,000$ random partitions shared by all MoM-based procedures and reused across outcomes ($\delta = 0.05$). For each outcome, the top rows give the dispersion of the partition-dependent point estimates (the partition-free Wald estimate in the label); the bottom rows give rejection and shape frequencies of the 95% confidence sets. The standard AR test is partition-free and shown for reference.

Outcome	Estimator	Median	SD	$Q_{0.05}$	$Q_{0.95}$
Worked for pay (Wald -0.1318)	Median-of-Ratios	-0.1335	0.0184	-0.1641	-0.1028
	Ratio-of-Medians	-0.1321	0.0189	-0.1638	-0.1016
	Catoni ratio	-0.1318	0.0000	-0.1318	-0.1318
Weeks worked (Wald -6.3596)	Median-of-Ratios	-6.3872	0.8206	-7.7276	-5.0832
	Ratio-of-Medians	-6.4020	0.8753	-7.8407	-4.9402
	Catoni ratio	-6.3595	0.0000	-6.3595	-6.3595
Hours/week (Wald -5.1872)	Median-of-Ratios	-5.1843	0.6893	-6.3345	-4.0640
	Ratio-of-Medians	-5.2561	0.7021	-6.4069	-4.0759
	Catoni ratio	-5.1871	0.0000	-5.1871	-5.1871
Labor income (Wald -2188.3)	Median-of-Ratios	-2192.4	395.1	-2833.3	-1546.7
	Ratio-of-Medians	-2171.4	398.9	-2823.9	-1526.5
	Catoni ratio	-2188.4	0.0	-2188.4	-2188.4
ln(Family income) (Wald -0.0262)	Median-of-Ratios	-0.0259	0.0469	-0.1032	0.0497
	Ratio-of-Medians	-0.0248	0.0487	-0.1032	0.0559
	Catoni ratio	-0.0262	0.0000	-0.0262	-0.0262

Outcome	Test	% rej. 0	% rej. β_{OLS}	% split	% unb.	Med. len.
Worked for pay	MoM-AR (feasible)	3.5	0.0	0.0	0.0	0.469
	SN-AR	99.3	1.2	0.0	0.0	0.144
	AR (standard)	100.0	0.0	0.0	0.0	0.103
Weeks worked	MoM-AR (feasible)	7.5	0.0	0.0	0.0	20.976
	SN-AR	99.6	1.4	0.0	0.0	6.366
	AR (standard)	100.0	0.0	0.0	0.0	4.636
Hours/week	MoM-AR (feasible)	6.0	0.0	0.0	0.0	17.710
	SN-AR	99.0	5.9	0.0	0.0	5.530
	AR (standard)	100.0	0.0	0.0	0.0	3.948
Labor income	MoM-AR (feasible)	1.1	0.0	0.0	0.0	10077
	SN-AR	90.4	1.2	0.0	0.0	3115
	AR (standard)	100.0	0.0	0.0	0.0	2251
ln(Family income)	MoM-AR (feasible)	0.0	0.0	0.0	0.0	1.233
	SN-AR	1.2	18.2	0.0	0.0	0.383
	AR (standard)	0.0	100.0	0.0	0.0	0.270

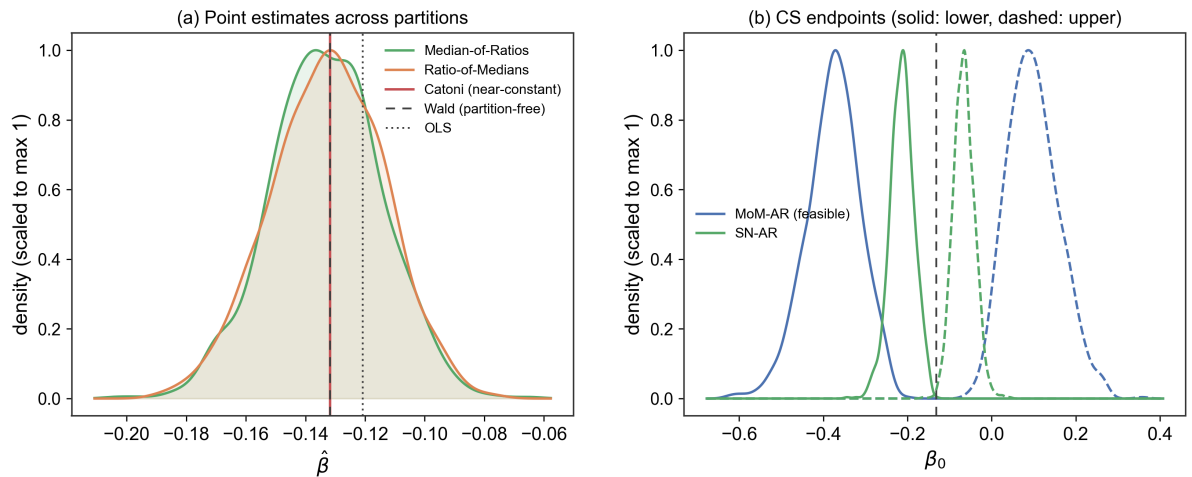


Figure 15: Partition sensitivity for Worked for pay, Angrist and Evans (1998) 1980 all-women sample, $B = 1,000$ random partitions. Panels as in Figure 14.

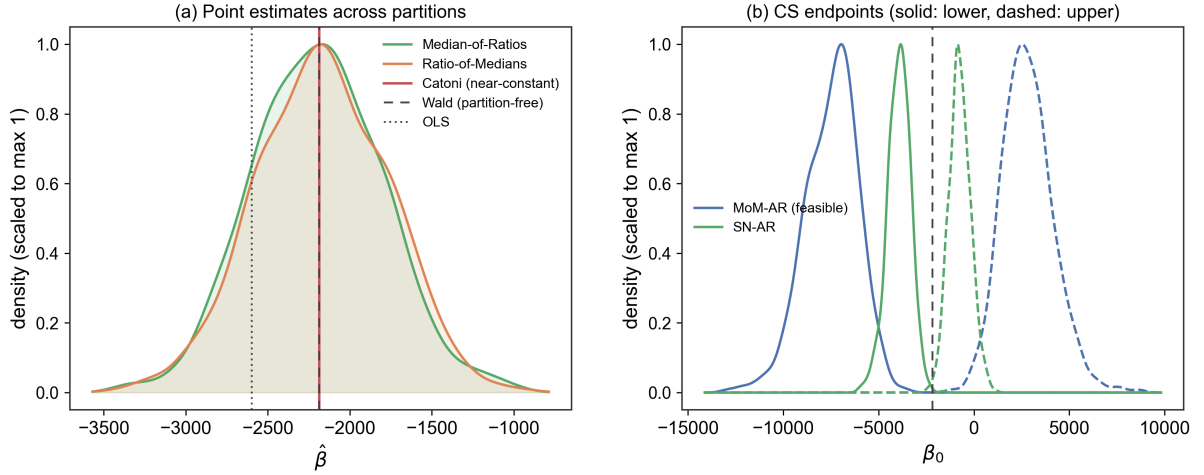


Figure 16: Partition sensitivity for Labour income, Angrist and Evans (1998) 1980 all-women sample, $B = 1,000$ random partitions. Panels as in Figure 14.

Expand: Here every aggregate lands on the partition-free Wald value: over the five outcomes the largest gap is 0.4%, Weeks worked at -6.3872 against -6.3596 . That settles the reading offered for Table 6, where on one partition Median-of-Ratios sat 21% below the Wald estimate for Labour income and Ratio-of-Medians 31% above it, and Ratio-of-Medians even reversed sign for $\ln(\text{Family income})$ at $+0.0481$. Aggregated, those become -2192.4 and -2171.4 against -2188.3 , and -0.0248 against -0.0262 : the gaps were the draw of the partition, not the estimator, and aggregation removes them. The verdicts do not move, the self-normalised set excluding zero for the four labour-supply outcomes and containing it for $\ln(\text{Family income})$, which is what Section 9.3.4 anticipated in finding every partition already unanimous; what is new is that the reported numbers no longer depend on a seed. As in Section 9.2.5 the aggregated sets have close to the median per-partition length for every outcome and every test, and the feasible MoM-AR set still contains zero throughout.

10 Discussion

Write discussion and conclusion

Table 8: Aggregated estimates and aggregated 95% confidence sets on the AE98 1980 all-women sample, over all $B = 1,000$ random partitions of Table 7, shared across outcomes (Definition 6.1 and Corollary 7.19, $\delta = 0.05$). Table 6 is the same quantities on a single partition. The Wald estimator and the standard AR test do not depend on the partition and are shown for reference; by Corollary 7.19 the aggregated sets are guaranteed at 2δ , not δ .

Outcome	Wald / Mean IV	Median-of-Ratios	Ratio-of-Medians
Worked for pay	-0.1318	-0.1335	-0.1321
Weeks worked	-6.3596	-6.3872	-6.4020
Hours/week	-5.1872	-5.1843	-5.2561
Labor income	-2188.3	-2192.4	-2171.4
ln(Family income)	-0.0262	-0.0259	-0.0248
Outcome	MoM-AR (feasible)	SN-AR	AR (standard)
Worked for pay	$[-0.374, 0.095]$	$[-0.213, -0.067]$	$[-0.184, -0.080]$
Weeks worked	$[-17.18, 3.79]$	$[-9.89, -3.48]$	$[-8.68, -4.04]$
Hours/week	$[-14.28, 3.59]$	$[-8.22, -2.69]$	$[-7.16, -3.22]$
Labor income	$[-7319, 2722]$	$[-3899, -772]$	$[-3313, -1061]$
ln(Family income)	$[-0.661, 0.583]$	$[-0.234, 0.152]$	$[-0.161, 0.109]$

11 Appendix

11.1 Self-Normalised Anderson-Rubin Test table

Table 9: Simulated critical values $c_{k,\delta}$: the $(1 - \delta)$ quantiles of R_k (Proposition 7.13).

k	$\delta = 0.1$	$\delta = 0.05$	$\delta = 0.025$	$\delta = 0.01$
5	2.773	4.236	6.275	10.300
10	1.569	2.085	2.672	3.614
15	0.938	1.181	1.438	1.798
19	0.800	0.996	1.190	1.459
20	0.846	1.052	1.259	1.540
24	0.743	0.916	1.086	1.311
25	0.673	0.828	0.980	1.177
30	0.640	0.784	0.920	1.100
37	0.535	0.651	0.761	0.902
40	0.535	0.649	0.758	0.895
50	0.469	0.567	0.659	0.772

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